AN ATTENTION BASED CNN-TRANSFORMER NETWORK FOR ECG BEAT CLASSIFICATION USING MULTI-LEAD SIGNALS

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CERTIFICATE

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PROJECT SUMMARY

S.No	Item	Description
1	Project Title	An Attention Based CNN-Transformer Network for ECG Beat Classification Using Multi-Lead Signals
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4	Research Group	Medical signal processing, Deep learning
5	Application Area	Health care
6	Aim of the Project	To develop an automated system that accurately classifies ECG heartbeats into different disease categories using multi-lead signals.
7	Project Outcomes	The model achieved good accuracy on standard dataset and proved effective for faster, reliable detection of heart abnormalities, showing potential for real-time healthcare applications.

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ABSTRACT

Electrocardiogram (ECG) signals are of primary importance in the diagnosis of cardiovascular diseases, which are still one of the primary causes of death worldwide. However, it is a complicated and time-consuming process of ECG signal analysis that very much relies on the knowledge and experience of an expert. Therefore, there is the urgent need to come up with effective automated ECG diagnosis systems. This paper will offer a new architecture in automated ECG signal classification pattern based on deep learning approach. The given framework starts with the beat segmentation of the ECG signals in which the R-peak can be identified and then the normalized heartbeat segments can be captured out of the twelve-lead ECG signals. This method is efficient for extracting the spatial and temporal characteristics of ECG signals. Next, a hybrid model of Convolutional Neural Networks (CNN) and transformer encoder layers was utilized for efficient ECG beat classification features. The convolutional layers are efficient in capturing the local features of ECG beat morphology, and the transformer layer is efficient in capturing the long-range temporal dependencies of ECG signals. In addition, a temporal attention mechanism is included in the model to emphasize the clinically important regions of the heartbeat while performing the feature extraction task. Finally, the extracted features are used with fully connected layers to perform the classification task. The proposed framework was validated using the PTB-XL dataset and focuses on major cardiovascular diseases, including Conduction Disturbance (CD), hypertrophy (HYP), Myocardial Infarction (MI), normal (NORM), and ST/T Changes (STTC). Experimental analysis validates the proposed model in the context of ECG beat classification, which has the potential to support intelligent decision-making.

Keywords: Electrocardiogram (ECG), ECG beat classification, Convolutional Neural Networks (CNNs), Transformer encoder, Temporal attention mechanism

CHAPTER-1

Introduction

1.1 Origin of the problem

The electrocardiogram (ECG) beat classification problem is linked with the growing amount of cardiovascular diseases on the planet. Cardiovascular conditions such as arrhythmia, myocardial infarction, hypertrophy, and conduction problems have become serious factors leading to premature deaths and morbidity. It is crucial to diagnose and treat the mentioned disorders in order to prevent heart conditions.

Cardio electric phenomena were studied since the end of the nineteenth century as the most effective method of non-invasive diagnostics. The modernization of electrocardiograms began in 1903 when the string galvanometer was invented by Willem Einthoven. From analog instruments to a paper strip to a multi-lead monitoring system in hospitals, the ECG underwent significant changes over time.

Typically, electrocardiographic diagnosis is done manually by cardiologists or other trained specialists. In manual diagnosis, a specialist analyzes waveforms and intervals (P, QRS, and T waves; PR, QT, and RR intervals), and detects disorders based on visual data and his experience. Nevertheless, with the rise in the amount of patients requiring examinations and the necessity to analyze ECGs lasting for a long period of time, certain problems emerged.

This issue stemmed primarily from the fact that manual ECG analysis requires a considerable amount of time and work, as well as expertise. Even trained specialists sometimes struggle to conduct an analysis of extensive ECG readings, particularly in situations where there is an unidentifiable wave pattern anomaly or a long period of ECG monitoring[11]. In addition, individual interpretations of ECG findings may vary between practitioners, making the process less consistent.

The same challenge emerged in single-lead ECG machines since these devices failed to capture all the electric activities of the heart. Certain cardiac conditions could become apparent in certain leads only, but traditional devices often could not detect all aspects of heart function and failed to produce accurate results.

The importance of this issue increased significantly due to the widespread adoption of digital healthcare systems, sensors, telemedicine, and remote patient monitoring services. All of those innovations produce an enormous number of ECG records each day, which cannot possibly be interpreted manually by humans. This is why experts had to look for a solution in automation, starting to develop automated ECG interpretation software back in the 1960s and 1970s.

Recently, deep learning methodologies including Convolutional Neural Networks (CNNs), Transformer Encoder, and Attention Mechanism have demonstrated impressive proficiency in feature extraction directly from the ECG signals to identify specific characteristics of their waveform. The aforementioned approaches enable identification of not only beat morphology but also long-range temporal dependence, thereby providing significant potential for the application to multi-class ECG beat classification.

Accordingly, the reason for the emergence of the current challenge is related to the inherent shortcomings of conventional ECG analysis, including manual analysis and the usage of a single lead, in addition to growing requirements for rapid and high-performance automatic classification of cardiac diseases.

Cardiovascular diseases (CVDs) are one of the health issues that have risen to become a serious problem in all parts of the world, and thus the use of effective diagnostic instruments is necessary. Some of the simplest tools employed in the diagnosis of heart related issues include electrocardiogram (ECG), which is used to measure the electrical activity of the heart in order to identify a range of heart rhythm abnormalities, including cardiac arrhythmias. Human analysis of ECG signals although a traditional one is a slow, laborious and subjective task particularly when dealing with irregular heartbeats not very frequent nor pronounced[1,2]. This issue is more urgent in the light of the growing amount

of long-term ECG records, and so, there is a strong necessity to make automated, reliable and highly accurate classifiers.

Major progress has been achieved in the recent past by introduction of deep learning paradigms which have increased the capability of medical diagnostic devices to interpret multifaceted physiological data significantly. It has been shown that CNNs are useful in extracting features in ECGs and other time-series data through learning the complex patterns in the raw data[3,4]. The latest release of the Transformer model, which has been applied to natural language processing, has also penetrated the processing of different kinds of data, impressively performing on the analysis of long-range dependencies of sequential data. It has also been discovered that the incorporation of an attention element into the network has increased the concentration of the network on the essential aspects of the ECG signal.

An elaborate analysis of the heart functions usually involves interpretation of different ECG signal leads, which provides a more elaborate account of heart functions as opposed to one lead signal[8,9]. It is possible that with appropriate use of various lead signals, the effectiveness of classification model discriminability can be significantly enhanced. Nevertheless, the most significant challenges (although these methods have been improved) are the creation of a model that may be effective in detecting local features and global time-varying dependencies of different ECG signal leads and, in addition, understand significant portions of the signal.

The scope of this document will be to provide a literature review of applying an attention-based CNN-Transformer model to ECG beat classification, which in this case is based on different lead signals. The effectiveness of CNNs, Transformers, and attention methods together can significantly enhance the classification of ECG signals by means of a new model. This paper will set out to give a clear background and state of the art and theoretical foundation that underlines the effectiveness of such a model in enhancing ECG signal classification.

1.2 Basic Definitions and Background

Before knowing the proposed methodology for classifying the ECG beats, some essential terminology and definitions associated with the project should be known.

1.2.1 Basic Definitions

Electrocardiogram (ECG):

Electrocardiogram (ECG) is a clinical examination that captures the electrical activity of the heart within a certain period of time. Heart generates small electrical signals during its contraction and relaxation processes. These signals are collected using electrodes attached to the chest, limbs, and feet. They are then represented in wave forms.

ECG plays an important role in diagnosing various cardiac conditions, such as heart rhythm disorders, myocardial infarction, hypertrophy, conduction disorders, etc. Due to its noninvasive, inexpensive, and informative nature, it is the most commonly used procedure in hospitals.

ECG Wave Components

The following three major elements are observed in a normal ECG waveform pattern:

P Wave: It shows the electrical signal generated due to the contraction of the atrium.

QRS Complex: It is the electrical signal produced by ventricular contraction and the most important element of the ECG process.

T Wave: It indicates the electrical activity during ventricular repolarization.

ECG = P + QRS + T

This waveform information helps in diagnosing the normal and abnormal functions of the heart.

R-Peak Detection

R-peak represents the peak in QRS complex and it is one of the most critical peaks in the ECG signal analysis process. The detection of the R-peak is critical as it plays a major role in detecting individual beats and performing beat segmentation in the ECG signal analysis process.

Beat Segmentation

ECG beat segmentation is a technique that involves dividing a continuous ECG signal into several segments, each containing an individual heartbeat. These beats contain information needed to perform classification. Beat segmentation is essential in the ECG signal analysis process as machine learning and deep learning algorithms work effectively with fixed-size beats.

Multi-Lead ECG

A multi-lead ECG system is designed to detect heart electrical activity from several directions using different leads. A typical clinical ECG uses 12 leads to give a full spatial representation of heart activity.

Unlike a single lead ECG, the multi-lead ECG system is more detailed since some abnormalities can only be observed in certain leads.

CNN (Convolutional Neural Network)

Convolutional Neural Network (CNN) is a type of deep learning algorithm used primarily for automatic feature extraction. CNN is applied in this project to identify vital features from the ECG signal waveforms such as peak shape, waveform, and local temporal variation.

A Transformer Encoder

Transformer Encoder is an advanced deep learning architecture that detects long-range temporal correlations within sequences. Since ECG data form a sequence, the Transformer helps understand relationships between different time points of the cardiac cycle.

Attention Mechanism

The Attention layer helps the model to pay attention to significant areas in the ECG data. Instead of treating all parts of the waveform equally, attention highlights the clinically important portions like abnormalities in the wave or changes in ST-T waves.

1.2.2 Background of the study

The analysis of electrocardiogram (ECG) signals is important in early diagnosis and detection of cardiovascular diseases, which are some of the main causes of deaths all over the world. Conventionally, ECG has been interpreted by hand by trained cardiologists, who have used their knowledge to detect irregularities in the pattern of waves. Manual analysis is however costly in time, subjective and it is also liable to human error particularly when addressing extensive data. This has prompted increased desire to have automated ECG classification systems that can help healthcare professionals make quicker and more precise diagnosis.

Machine learning and deep learning methods have been popularly used in ECG signal classification in recent years. Conventional machine learning models, including Support Vector Machines (SVM) and k-Nearest Neighbors (KNN), depend on handcrafted feature selection, need domain expertise and do not provide an effective way of capturing a complex pattern. As deep learning has progressed, it has demonstrated vast improvement in models such as Convolutional Neural Networks (CNN) and Recurrent Neural Networks (RNN) when these models are left to learn features directly by looking at raw ECG signals. CNNs have the ability to extract morphological structures locally, while RNNs and LSTM models can be applied to extract temporal dependencies in sequential data.

A recent trend has been to explore more complex models such as Transformers and Attention Models for the ability to extract long-term dependencies and focus on important signal components. These methods address the shortcomings of the traditional sequence models, relying on self-attention mechanisms that can process the whole signal at once. Also, the performance of the classification has been enhanced due to the use of multi-lead ECG data that give a more detailed picture of the cardiac activity. The advancements have led to a new direction of hybrid models integrating CNN, Transformer, and Attention mechanisms that can be used to achieve more accurate and robust ECG classification systems.

1.3 Problem Statement with Objectives and Outcomes

1.3.1 Problem Statement

Heart diseases have become one of the leading causes of deaths all over the world. Early and precise diagnosis has become essential due to this situation. Electrocardiogram (ECG) is one of the common diagnostic techniques that help to diagnose the irregularities in electrical signals of the heart. The manual method of interpretation of ECG data by cardiologists is usually time-consuming, costly, and requires extensive expertise.

The increasing number of patients and the continuous availability of medical services in healthcare facilities have led to the emergence of a large volume of ECG recordings every day. Manual examination of this massive data is cumbersome and may cause delays in diagnosing the disease. Also, current automated systems that rely on handcrafted features cannot analyze complex ECG waveforms accurately.

Single-lead ECG is another limitation in many existing approaches since they are unable to provide comprehensive information regarding the cardiac signals of patients. Myocardial infarction, conductive disorders, and ST-T alterations are some cardiac abnormalities that show up differently in several leads. Therefore, single-lead systems are less reliable compared to other approaches.

The issue that will be solved through the proposed project relates to the necessity for the development of an AI-based automatic beat classification system that will be able to classify the input multi-lead ECG signals into different types of cardiac diseases. It should be able to learn about the local features as well as the long-term relations of the signals within the ECG beats.

This project seeks to solve the above-mentioned issue by building an attention-based convolutional neural network model with transformer encoder on multi-lead ECGs from the PTB-XL database.

1.3.2 Objectives

The primary objectives of the study are:

- Data acquisition and preprocessing for multi-channel ECG signals from PTB-XL database.

- Perform R-peak detection and heart beat segmentation to obtain individual beats.
- Normalize ECG beats to achieve constant deep learning training.
- To propose a CNN architecture as a feature extractor for ECG waveform morphologies.
- To apply Transformer Encoder to model temporal dependencies within ECG beats.
- To implement attention mechanism to focus on the critical regions of ECG signals.
- To classify ECG beats into five categories: CD, HYP, MI, NORM, STTC
- Evaluate model performance based on accuracy, confusion matrix, ROC curves, precision, recall, and F1-score.

1.3.3 Outcomes

The key outcomes obtained during the implementation of the research include:

- Successful extraction of beat-wise ECG data from multi-channel PTB-XL data.
- Development of deep learning model based on attention-based CNN + Transformer encoder.
- High classification accuracy rate of 97.06%.
- Model performance analysis based on:
  - Accuracy and loss graphs
  - Confusion matrix
  - ROC-AUC curves
- Enhancement of multi-class ECG beats classification.
- Demonstration of multi-lead deep learning-based cardiac diagnostic system efficiency.

The obtained result indicates that the proposed model can be applied in automatic and effective detection of cardiac disorders, which will help doctors in fast diagnosis and treatment of cardiac patients.

1.4 Related Applications

The potential uses of the proposed Attention-based Convolutional Neural Network for ECG Beat Classification Using Multi-Lead Signals in the realm of medicine and

bioinformatics are numerous. Since the Electrocardiogram (ECG) is commonly used by medical practitioners in diagnosing the condition of the heart of a patient, there could be many instances wherein its classification can prove useful.

1. **Hospital Cardiac Monitoring System**

   Firstly, the model can be applied to cardiac monitoring systems operated by hospitals. Continuous monitoring of patients is done in ICU, emergency units, cardiological departments of hospitals via ECG devices. The suggested algorithm could automatically classify ECG beats, which would help cardiologists detect myocardial infarction, hypertrophy, conduction disturbances, and ST-T changes.

   This way the burden will be decreased, and the decision-making process accelerated.

2. **Wearable Health Monitoring Devices**

   In the current era, the use of wearables such as smartwatches, portable heart signal analyzers and IoT-enabled devices, provides continuous monitoring of patients' vital signs. This results in large-scale data sets that need to be analyzed immediately.

   The developed model can help with providing:

   - Early warnings
   - Continuous heart monitoring
   - Heart health tracking
   - Instant abnormal heartbeat detection

   Such solutions can benefit elderly patients and those with chronic heart diseases significantly.

3. **Telemedicine and Distance Healthcare**

   As more telemedicine services become popular, many patients get consultations using telemedicine facilities. In such cases, ECG data obtained from patients via remote locations could be automatically classified and delivered to doctors.

Such approach will allow healthcare specialists to obtain a better understanding of the patient's health situation fast.

4. **Clinical Decision Support Systems**

The suggested ECG classification algorithm could serve as a component of a Clinical Decision Support System (CDSS). Such software will help cardiologists obtain automatic suggestions on diagnosis, based on the analysis of ECG data.

This approach allows minimizing possible mistakes that humans might make when interpreting the waves.

5. **Medical Research and Biomedical Engineering**

Automatic ECG classification algorithms are very useful for researchers studying diseases related to cardiac disorders. Researchers working on cardiac abnormalities can take advantage of this algorithm to conduct biomedical research, including:

- Massive ECG signal analysis
- Pattern detection
- Trends of particular diseases detection
- Studies involving multi-class cardiac abnormalities.

Usage of multi lead deep learning systems improves the biomedical research.

6. **Emergency Alert Systems**

The approach can also be used in real-time systems to detect emergency situations where urgent attention is required. Early detection of cardiac emergencies, for instance, by automatically classifying electrocardiograms can provide an emergency warning for immediate action. This would be useful in reducing fatalities from such cases.

CHAPTER-2

Review of Literature

2.1 Description of Existing systems

Cardiovascular diseases (CVDs) are one of the health issues that have risen to become a serious problem in all parts of the world, and thus the use of effective diagnostic instruments is necessary. Some of the simplest tools employed in the diagnosis of heart related issues include electrocardiogram (ECG), which is used to measure the electrical activity of the heart in order to identify a range of heart rhythm abnormalities, including cardiac arrhythmias[1,2]. Human analysis of ECG signals although a traditional one is a slow, laborious and subjective task particularly when dealing with irregular heartbeats not very frequent nor pronounced[1,2]. This issue is more urgent in the light of the growing amount of long-term ECG records, and so, there is a strong necessity to make automated, reliable and highly accurate classifiers.

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The classification of the electrocardiogram (ECG) beats has experienced remarkable advancements, which have been driven by the need to have an automatic and accurate classification of the arrhythmia. The first approaches were mostly based on conventional machine learning techniques with a focus on feature engineering. Approaches that relied on Support Vector Machines (SVM) and K-Nearest Neighbors (KNN) were often associated with complex preprocessing and feature design based on ECG morphology or statistics[10,12]. An example is a study that investigated the use of tunable Q wavelet transform and statistical feature extractors along with SVM with an accuracy of 97.30%[2]. These techniques gave workable accuracy on particular datasets, but their overall stability depended on the variability of the ECG signals, and the high manual effort required in feature extraction[4].

With the emergence of deep learning, most of the drawbacks of manual feature engineering were reduced. Convolutional Neural Networks (CNNs) proved to be especially useful when analyzing ECG signal because of their capability to automatically extract hierarchical signals off unprocessed time-series data. The use of one-dimensional (1D) CNNs has been popular because they combine both morphological and temporal features of cardiac cycles[13]. As an example, a 1D-CNN model that uses RR interval features and entropy rate was able to achieve average accuracy of 97.91% on inter-patient classification of the MIT-

BIH dataset[3]. A nine-layer deep CNN model was also found to automatically classify ECG heartbeats into five categories with high accuracy and without explicitly extracting features[10]. Hybrid CNNs, which interoperate CNN blocks with Recurrent Neural Networks (RNNs) (e.g. Long Short-Term Memory or Gated Recurrent Unit) were also popularized due to the sequential dependencies that they are able to models. Similar models such as the 2D-CNN-LSTM transformed 1D ECG signals into 2D scalogram images to enable effective classification of the different cardiac disorders[2]. On the same note, CNN-BiLSTM models, extended to include attention controls have established a high level of performance on the MIT-BIH arrhythmia database through the use of adaptive synthetic sampling to deal with the issue of class imbalance[14].

Transformer architecture After its original application in natural language processing, the Transformer architecture has proven to be remarkably effective in sequence modeling and has been applied to other areas of application, such as time-series analysis in bioinformatics and genome data[5,6]. Transformer encoders to classify heartbeats have been modified to allow ECG analysis, where it is known that the ECG signal has a time-sequence characteristic. Transformer models have demonstrated good results in the ability to capture rhythmic patterns in ECG signals[15]. Nonetheless, the application of Transformers directly to long sequences of ECGs may be very costly in computation. To overcome this, refined attentions including refined diagnosis and gated linear (GAL) attentions have been suggested in order to reduce the computational stresses whilst not compromising the performance so much when detecting arrhythmias[16]. Transformer based architectures such as Beat-aligned Transformer (BaT) utilize the cyclicity of ECG signals successfully by multiplexing the input as a series of synchronized beats and show competitive performance with other state of the art approaches[17]. ECG former model also uses Transformer capabilities to classify arrhythmia using various datasets[18].

Attention models have been merged with other deep learning architectures to improve the capacity of the model to concentrate on informative sections of the signal input. Selective attention processes may also be used in ECG classification to emphasize key areas and ignore undesired noise to enhance diagnostic accuracy by a large margin[7]. In hybrid models, multi-head attention can be used, which enables taking into account various

representations of input data, which enhances the classification. An example is a multi-module Recurrent Convolutional Neural Network (RCNN) comprising of a Transformer encoder and multi-head attention that obtained good results in ECG classification by integrating time-sequences, spectrogram, and metadata modules[19]. On the same note, a hybrid CNN-BiLSTM-BiGRU algorithm that was aided by a multi-head self-attention mechanism achieved high classification accuracy using single-lead ECG data[20]. The explainability provided by attention maps which emphasize on the areas of much influence of the signal also supports clinicians in reasoning with models and making well-informed diagnosis decisions.

Multi-lead ECG signal use is instrumental in the achievement of a clear picture of cardiac activity. Most studies are concerned with single-lead classification, but multi-lead processing models have the potential to provide more diagnostic data. Multi-lead processing was found useful as a wide and deep Transformer neural network that classified 12-lead ECG sequences into multiple classes of cardiac abnormality scored highest at the PhysioNet challenge[8]. A second novel deep learning model that involves a hybrid of CNNs and channel attention mechanism was successfully applied to recognize arrhythmia using both 2-lead and 12-lead ECG signals[9]. The presented integration of multi-lead into novel architecture CNN-Transformers with attention is a breakthrough towards more accurate and clinically relevant automated ECG diagnostics.

2.2 Summary of Literature Study

Table 2.2: Summary of literature review

S. NO	Paper & Author name	Year of publication	Algorithms	Workdone	Conclusion
1	Interpreting Deep Neural Networks for Single-Lead ECG Arrhythmia	2020	Deep Neural Networks (DNN) CNN-based models	The authors created a deep learning-based architecture for arrhythmia classification based on single-lead ECG	According to the researchers' findings, deep neural networks prove to be highly efficient when

	Classification Vijayarangan, S. et al.		Model interpretation techniques (like saliency/attention visualization)	records. The work mainly emphasized not only obtaining good results in classification but also making deep neural networks more interpretable. Convolutional neural network algorithms were utilized to obtain relevant features from ECG data, while the interpretation of the model was done using visualization approaches to determine which sections of the input signal influenced the model's decision the most.	performing arrhythmia classification based on single-lead ECG records. Moreover, applying interpretability approaches makes the use of deep neural networks more reliable in the clinical setting.
2	A Hybrid Deep Learning Approach for ECG-Based Arrhythmia Classification Madan, P. et al.	2022	Hybrid Deep Learning: CNN (for feature extraction) RNN / LSTM (for temporal dependencies)	The authors suggested a hybrid deep learning model based on convolutional neural networks (CNN) and long short-term memory (LSTM) networks to classify arrhythmia during ECG. Spatial features of ECG signals were automatically selected using CNN, and the temporal dependencies of the sequence were learned with LSTM. This model was	The researchers came to the conclusion that hybrid deep learning techniques are much effective in terms of classification performance in comparison with single models. CNN and LSTM are effective in the combination of their ability to capture the spatial and

				trained and tested on ECG datasets to classify various types of arrhythmias, which reveals the benefits of using feature extraction and sequence modeling methods.	temporal features of the ECG signals resulting in increased accuracy and strength. This method is especially applicable to time-series medical data such as ECG.
3	Arrhythmia Detection in Inter-Patient ECG Signals Using Entropy Rate Features and RR Intervals with CNN Architecture Berrahou, N. et al.	2024	CNN (Convolutional Neural Network) Feature Engineering: Entropy rate features RR interval features	This paper is concentrated on detecting the arrhythmia with inter-patient ECG signals with a combination of handcrafted features and deep learning. The authors used entropy-based features and RR interval features to capture variability and complexity of heart rhythms. These attributes were then inputted into a convolutional neural network (CNN) to be classified. The paper had a focus on the management of inter-patient variability so that the model can be effective on various individuals.	The authors found that the combination of handcrafted attributes like entropy and RR intervals with CNN enhances classification and generalization between patients. The model showed good results in the way it managed differences among various individuals, and it is more applicable to real-world clinical practice.

4	A Historical Survey of Advances in Transformer Architectures Sajun, A.R. et al.	2024	Transformer Architecture Self-Attention Mechanism Variants of Transformers	The authors had an elaborate survey of the development of Transformer architectures, their development and their usage in sequence modeling tasks. The article provided the mechanism of self-attention and how Transformers can reach the shortcomings of the more traditional sequence models, such as RNNs. It also discussed different variants of Transformers and their implementation in other fields, such as in healthcare and time-series data analysis.	This paper has found that Transformer architectures are well suited to prediction of sequential data because they can address long range dependencies with self-attention based mechanisms. They perform better than traditional models in a wide range of sequence-based tasks, and are also finding more and more uses in medical signal processing applications, including ECG analysis.
5	ECG Signal Classification Review Fikri, M.R. et al.	2021	Machine Learning (SVM, KNN) Deep Learning (CNN, RNN)	The paper has introduced an in-depth survey of the ECG signal classification algorithms, including the classical machine learning algorithms and the latest deep learning algorithms. The authors examined different algorithms	The review showed that the deep learning techniques, especially CNN and RNN-based models are better at ECGs classification tasks than conventional machine learning

				including support vector machines, k-nearest neighbors, convolutional neural networks and recurrent neural networks. They have also addressed various preprocessing, datasets and evaluation methods of the ECG classification studies.	techniques. Nevertheless, there are still challenges like data quality, computational complexity, and implementation requirements that are required to be real-time. The research has emphasized the need to come up with more effective and precise models that can be used in the clinical context.

CHAPTER-3

Proposed Method

3.1 Design Methodology

The suggested system will incorporate convolutional neural networks, Transformer encoders, and a temporal attention system to be successful in capturing both the short-term morphological and long-term temporal dependencies of ECG signals. The entire architecture is made up of five key steps that include: ECG data preprocessing, beat segmentation, feature extraction via convolutional neural networks, temporal dependency modeling via a Transformer encoder, and final classification via fully connected layers.

3.1.1 ECG Dataset and Preprocessing

The suggested approach provides the use of the PTB-XL dataset, which is a large publicly available electrocardiogram dataset that consists of 12-lead ECG recordings. All ECG records are sampled at a constant rate and labeled with diagnostic codes of various cardiac anomalies.

ECG signals are first preprocessed to have stable input representation before the model is trained. The metadata of the dataset is then processed in order to derive diagnostic classes on the present statements of SCP. Such diagnostic codes are coded to more generic diagnostic codes of various cardiac conditions. In order to prevent imbalance in classes during the training, a balanced dataset will be created by sampling an equal amount of ECG records of each diagnostic category.

Let the raw ECG signal be denoted as Eq. (1)

$$X = \{x_1, x_2, \dots, x_T\} \qquad (1)$$

where T is the total time samples in the signal.

3.1.2 R-Peak Detection and Beat Segmentation

R-peaks have to be correctly identified in time series ECGs in order to isolate R-peaks. R-peaks in this work are observed with the help of the peak detection algorithm applied to the second ECG lead (Lead II), which usually gives good visibility of the QRS complex.

Assume that the R-peaks in the ECG signal have been identified and denoted as p_i shown in Eq. (2):

$$P = \{p_1, p_2, \dots, p_n\} \qquad (2)$$

where n is the number of beatings that were detected.

The segments of every heartbeat are obtained by choosing a predefined window of the R-peak. In particular, 35 samples are cut out in front of the R-peak and 35 samples are cut out after the R-peak, which gives a beat length of 70 samples. The heartbeat is then extracted and normalized so as to minimize variation in amplitude of the heartbeat across the recordings as Eq. (3):

$$x_{norm} = \frac{x - \mu}{\sigma + \epsilon} \qquad (3)$$

in which μ and σ are the mean and standard deviation of the signal of the heartbeat, respectively, and ϵ is a small constant to ensure the numerical stability.

Every extracted heartbeat will be expressed as an M multi-lead matrix having the size: 12×70

In which, 12 means the number of ECG leads and 70 is the number of temporal samples of a heartbeat.

3.1.3 CNN-Based feature extraction

Convolutional Neural Networks (CNNs) are utilized in order to identify local morphological characteristics of ECG signals. Of interest is that CNN layers are especially useful in the recognition of typical ECG waveform patterns, namely, P waves, QRS complexes, and T waves.

Given an input ECG beat X is the convolution operation as Eq. (4):

$$h_i^l = f\left(\sum_{j=1}^{k} W_j^l * x_{i+j-1}^{l-1} + b^l\right) \qquad (4)$$

where:

h_i^l is the output feature map of layer l.

W^l is the convolution operator.

b^l is the bias term.

$f(\cdot)$ represents the nonlinear activation function.

k represents the kernel size.

To extract increased higher-level ECG representations step-wise, the proposed architecture employs 3 convolutional layers. Finally, every layer of the convolution is followed by a batch normalization and ReLU activation to stabilize training and enhance learning of features.

3.1.4 Transformer Encoder of Temporal Modeling

CNN layers are an effective method to capture local signal morphology though they might not be able to capture long-range temporal dependencies across ECG sequences. In order to overcome this shortcoming, a Transformer encoder is added to the convolutional feature extractor.

Transformer is a neural network that makes use of a self-attention mechanism to learn the relationship between the various time steps of the ECG signal. Given a series of feature vectors X, the self-attention mechanism calculates three representations:

$Q = XW_Q,\ K = XW_K, V = XW_V$

where Q, K and V are query, key and value matrices respectively, and W_Q, W_K and W_V are projection matrices, which can be learned.

The output on attention is calculated as Eq. (5):

$$Attention(Q, K, V) = softmax\left(\frac{QK^T}{\sqrt{d_k}}\right)V \quad (5)$$

where d_k is the dimensionality of the key vectors.

Multi-head attention enables the model to learn various patterns of relationships in temporal changes in ECG signals, since it computes multiple representations of attention and simultaneously.

3.1.5 Temporal Attention Pooling

Following the processing of the ECG sequence by the Transformer encoder, temporal attention mechanism is used to isolate the most informative parts of the heartbeat signal. This system ascribes greater significance to more important segments of ECG waveforms that play a major role in classification.

Attention weight of every time step is calculated as Eq. (6):

$$\alpha_t = \frac{\exp(W_\alpha h_t)}{\sum_{i=1}^{T} \exp(W_\alpha h_i)} \qquad (6)$$

where h_t is the hidden feature at time step t, and W_α is the attention weight matrix.

The last feature representation Eq. (7) is to find a weighted sum of all the temporal features:

$$z = \sum_{t=1}^{T} \alpha_t h_t \qquad (7)$$

This pooling based on attention enables the model to concentrate on clinically important parts of ECG at the expense of noise.

3.1.6 Classification Layer

The summed feature vector that is acquired of the attention layer is fed through fully connected layers to give the end classification. The softmax activation function is applied in the last layer to calculate the probability distribution of classes of the diagnosis as Eq. (8):

$$P(y = i|x) = \frac{e^{z_i}}{\sum_{j=1}^{c} e^{z_j}} \qquad (8)$$

where C is the number of diagnostic classes.

The model is trained with the help of cross-entropy loss function and the AdamW optimizer is used to perform optimization and early stopping is used to prevent the tendency to over fit the training.

3.2 System Architecture Diagram

Architecture of the proposed ECG beats classification model highlights the whole process of data flow within the model from the input of raw ECG data to the output of the final prediction for the ECG beats class. This helps show how different components within the architecture are interconnected so that the ECG signals can be efficiently classified.

This architecture is a hybrid approach consisting of signal pre-processing, CNN layer, Encoder transformer, attention pooling, and Fully Connected Classification Layers.

Here, **Figure 3.2.** shows the architecture diagram of the proposed model.

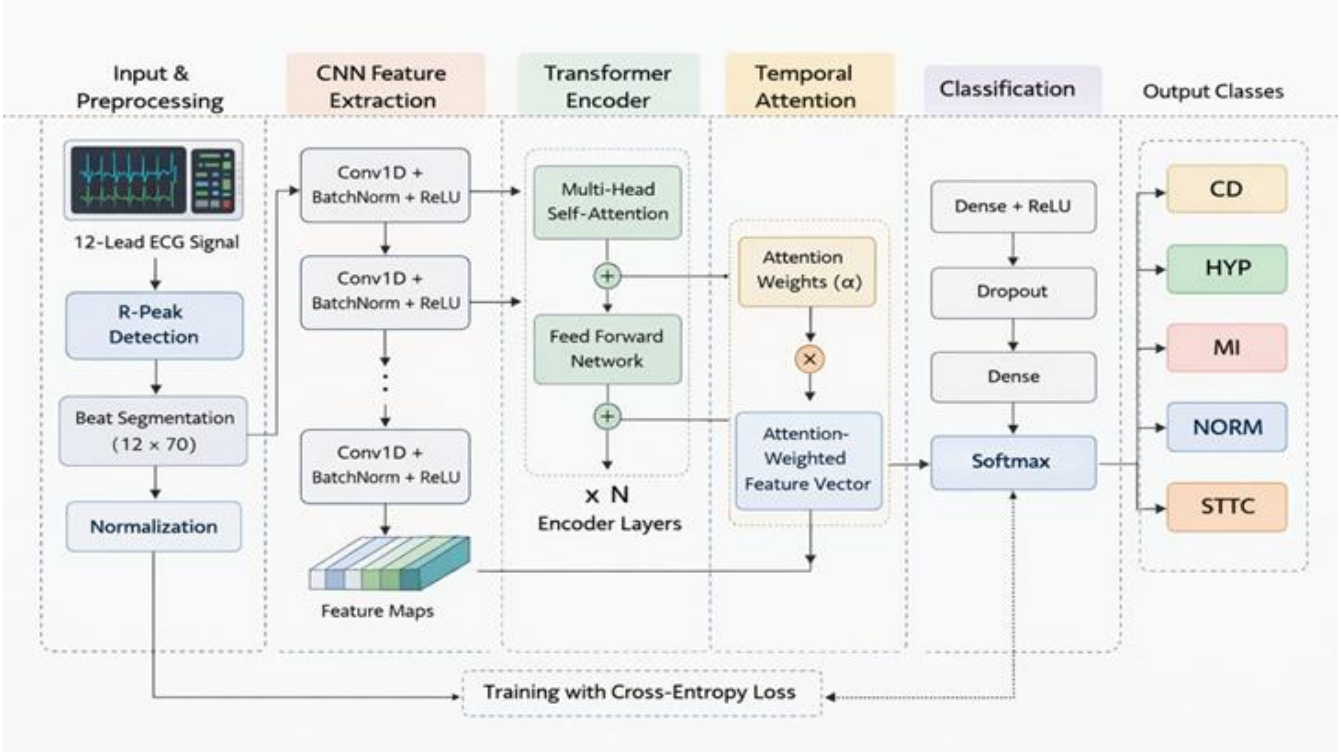

Figure 3.2: Architecture diagram

The general architecture is made up of the following key modules:

Input Layer

Input layer forms the very first component of the network architecture. The input layer receives the multi-lead ECG data taken from the PTB-XL database. Each and every ECG signal comprises 12 leads that help measure the heart's electrical activity in different directions, thus giving an overall idea about heart dynamics. Multi-lead signals provide more diagnostic data as compared to single-lead ECG, and therefore the model can identify subtle abnormalities that would not be seen with one lead of the signal. The R-peak-centered extraction is used to convert the continuous ECG signals into beat-level segments so that they can be used in learning. The most prominent point of the QRS complex is the R-peak and is identified and taken as a reference to isolate fixed length segments of heartbeat. It ensures that all input samples have relevance to a particular cardiac cycle, making the data more accurate and relevant for the model.

Preprocessing Module

Preprocessing module is essential in processing the raw ECG signals to enable effective learning. Because ECG signals are usually noisy because of muscle movement, electrode placement and external interference, noise filtering is used to get rid of any unwanted disturbance and improve the quality of the signal. After this, normalization is carried out to normalize the amplitude of ECG signals such that the differences in the scale of the signals do not influence the learning. Segmentation of the beat is then performed based on identified R-peaks with each ECG recording being segmented into smaller parts based on individual heartbeats. The conversion of long continuous signals into structured input samples is used as a step that makes the input signal more manageable by the model. Also, the label encoding is used to encode the categorical class labels into numerical format, which is compatible with the classification model. This is generally how the preprocessing module makes sure that the input data is clean, consistent and well structured, which contributes greatly to the performance and stability of the model.

CNN Feature Extraction Block

After segmentation, the ECG signals are passed to the preprocessing layer then through to the CNN feature extraction layer where the local morphological features of the ECG signals are learned. In the feature extraction layer, there are several Conv1D layers with batch normalization and ReLU activation layers. The Conv1D layers process the signal elements of the ECG in the time domain which allows the model to identify key pattern in the waveforms, including the QRS complex, P wave and T wave. Normalization of the outputs of every layer is done to stabilize and speed up the training process, whereas non-linearity is provided by the ReLU activation function that enables the network to learn more complicated patterns. Pooling layers are added to reduce dimensionality while preserving the important features. Hierarchical feature representation is learned in the CNN starting from the basic pattern like peaks and slopes and progressing to more complicated patterns like the wave morphology and segments variations through the stacking of the multiple Conv1D layers. The automatic feature extraction does not require manual feature engineering and creates a solid basis on further processing.

Transformer Encoder Block

The feature maps are generated by the CNN are then fed into the Transformer Encoder block, which increases the ability of the model to understand the time dependencies between the ECG signal. In contrast to CNNs that pay much attention to local trends, the Transformer Encoder considers long term dependencies by analyzing the entire sequence in parallel. This it does by a self-attention mechanism, with every timestep in sequence attending to every other timestep, which enables the model to learn contextual correlations between various components of the ECG signal. The encoder is composed of several layers, which are filled with multi-head self-attention and feed-forward neural networks, residual connections, and layer normalization to promote stable training. Multi-head attention allows the model to attend to various features of the signal simultaneously, enhancing its performance in capturing rich temporal features. This block is especially significant in ECG analysis where some cardiac abnormalities are based on interrelationships between remote waveform elements. The model uses the Transformer Encoder to give the signal a global perspective, and its local feature extraction through CNN.

Temporal Attention Pooling

After the Transformer Encoder, a temporal attention pooling process is implemented to further filter the features extracted. This layer gives varying weight to various time steps of the ECG signal and because of this the model can pay more attention to diagnostically important parts. Whereas in ECG signals, some of the elements like the QRS complex and the ST segment hold greater clinical value than others. What is obtained is the attention-weighted feature-vector that focuses on the significant patterns and enhances the discriminatory strength of the model. This step also increases the interpretability of the model, since it can give a clue to which components of the signal are the most important in the ultimate prediction. A temporal attention mechanism can significantly improve the ability of a neural network classifier since it allows it to pay attention to the necessary features only.

Softmax Output + Fully Connected Layers.

In the last layer of the model, the vector of the attended features goes through fully connected layers. These fully connected layers help convert the input features into a

representation that could be used for classification purposes. Dropout is used here to prevent overfitting. The last layer is a softmax activation, which transforms the output into the probability values of each of the classes. The model is based on five categories of ECG: Conduction Disturbance (CD), Hypertrophy (HYP), Myocardial Infarction (MI), Normal (NORM), and ST/T Changes (STTC), which the model predicts based on these probabilities. The model is optimized during the training by the cross-entropy loss that is the difference between true and predicted class labels and determines the learning process. The last classification phase combines all the learned features before to make the correct and dependable predictions and finally this completes the end-to-end ECG classification pipeline.

Working Principles of architecture

The architecture provides a description of the processing of the ECG signal by various layers of the deep learning model to obtain the correct heartbeat categorization.

The input of the raw multi-lead ECG signals in the PTB-XL dataset is the beginning of the process. The dataset includes 12-lead ECG recordings, which means that it gives various spatial images of the electrical activity of the heart. This assists the system to record the abnormalities which might manifest in only some leads.

The raw ECG signals are first filtered by the preprocessing block in the first stage. Here, undesirable noise, bias and signal interference are eliminated. The signals are then normalized to ensure that the range of values are consistent. R-peak detection follows after the preprocessing steps to determine the precise location of every heartbeat. The continuous ECG signal is divided into beat samples of fixed size using these R-peaks.

This is then fed to the CNN feature extraction block by giving the segmented beats. The CNN uses one-dimensional convolutional filters on the Ecg waveform in order to obtain significant local patterns including:

- QRS morphology
- peak amplitude

- waveform distortion
- ST segment variation
- local temporal structures

The features that each convolution layer learns are more and more complex features of the heartbeat. The pooling layers decrease the size of features, and important signal characteristics are retained.

The obtained feature maps are next fed to the Transformer Encoder block. The transformer, as opposed to CNN which predominantly concentrates on local patterns, captures long-range temporal relationships among various sections of the ECG signal. It takes self-attention to find out how a single point in the signal corresponds to another.
The features are sent to the Attention Pooling layer after undergoing transformer processing. This layer gives various weights of importance to various parts of the ECG signal. The weights placed on clinically significant regions of the waveform like abnormal QRS complexes, ST elevation, or T-wave changes are increased which enables the model to emphasize more on the information of interest in the diagnosis.

These weighted features are then fed into the refined weighted layer serving as the ultimate classifier. The output scores are transformed into probabilities of classes using the softmax activation function.

The system finally predicts one of the five ECG classes:

- CD
- HYP
- MI
- NORM
- STTC

The class of the predicted cardiac condition is the type of heartbeat condition that was detected on the input heartbeat.

Thus, the architecture works by combining:

- CNN local feature learning.
- global time perception with Transformer.
- verisimilitude and focus via Attention.

Such a combination of working principle enables the system to classify with high accuracy of 97.06%.

3.3 Description of Algorithms

PTB-XL Data Storage Layer:

- ECG recordings in .dat format
- Header files.hea
- Metadata files (ptbxl_database.csv, scp_statements.csv)
- It provides input to the Data Loader.

Data Loader & Preprocessing Module

- Load ECG signals using WFDB
- Normalize the signal
- Apply augmentation (noise addition, scaling, stretching)

Extract the following features:

- Wavelet coefficients
- FFT features
- Statistical features (mean, std)
- R-peak detection (from QRS complex)
- QRS variability

Following image conversion, the next step is edge detection. This is performed on the grayscale ECG images to highlight the structural components of the waveform. The Canny Edge Detection algorithm is applied to extract prominent features by identifying areas of rapid intensity change, which typically correspond to the P wave, QRS complex, and T wave in an ECG. The output of this step is a binary image where the edges are represented in white against a black background. This edge map retains the essential morphological

patterns of the ECG signal while discarding unnecessary background noise, thereby simplifying the data for the next phase.

3.3.1 Graph Construction

This module converts ECG signals into graph-structured data compatible with PyTorch Geometric.

Tasks:

- Treat each time sample as a node
- Assign extracted ECG features as node features
- Construct temporal edges using K-nearest neighbors (KNN)
- Build edge_index tensor for message passing
- Store graphs for training and evaluation

3.3.2 Graph Isomorphism Network (GIN)

- Graph Isomorphism Network (GIN) is a powerful Graph Neural Network architecture designed to distinguish graph structures with high precision.
- Construct graphs G=(V,E) from ECG signals with node features x_v.
- Initialize GIN with parameters Θ.
- Node update rule:

$$h_v^{(l)} = MLP^{(l)}((1 + \epsilon^{(l)}. h_v^{(l-1)} + \sum_{u \in \mathcal{N}(v)} h_u^{(l-1)}) \qquad (9)$$

- Graph readout:

$$h_G = READOUT(\left\{h_v^{(L)}\right\}) \qquad (10)$$

- Classification:

$$\hat{y} = softmax(Wh_G + b) \qquad (11)$$

- Training loop:
- For each epoch:
- Compute predictions, loss

$$L = -\frac{1}{b}\Sigma y log(\hat{y}) \quad (12)$$

- Backpropagate, update Θ.
- Return trained model M.

The Graph Isomorphism Network (GIN) is a powerful Graph Neural Network architecture designed to capture the structural relationships within graph-based data. It is theoretically proven to have the same discriminative capability as the Weisfeiler–Lehman (WL) graph isomorphism test, making it one of the strongest models for graph classification tasks.

3.3.3 Graph Attention Network(GAT)

- The Graph Attention Network (GAT) incorporates attention mechanisms into graph
- Processing, allowing nodes to assign different importance weights to their neighbors.
- Initialize GAT parameters Θ.
- Linear transformation:
- $h_v = Wx_v$ (13)
- Attention coefficients:

$$\boldsymbol{\alpha_{ij}} = \frac{\boldsymbol{exp}\left(\boldsymbol{LeakyReLU}(\boldsymbol{a^T}[\boldsymbol{Wh_i}\|\boldsymbol{Wh_j}])\right)}{\sum_{\boldsymbol{k}\in\mathcal{N}(\boldsymbol{i})}\boldsymbol{exp}(\boldsymbol{LeakyReLU}(\boldsymbol{a^T}[\boldsymbol{Wh_i}\|\boldsymbol{Wh_k}]))} \quad (14)$$

- Node update rule:

$$h_v^{'} = \sigma(\sum_{u\in N(v)} \alpha_{vu}^{m} W^{m} h_u) \quad (15)$$

- Multi-head attention
- Graph readout + classification:

$$\hat{y} = softmax(Wh_G + b) \quad (16)$$

- Training loop:
- For each epoch: compute predictions, loss, backprop, update Θ.

- Return trained model M.

The Graph Attention Network (GAT) introduces an attention mechanism into GNNs, allowing the model to learn which neighboring nodes are more important during message passing.

3.3.4 CNN for feature extraction

Convolutional Neural Network (CNN) is a deep learning algorithm that is used to automatically generate local features on input data through convolution. It is extensively applied in processing structured data such as images and time-series signals, such as ECG signals.

CNN is important in ECG classification project in extracting local morphological features of the ECG beats. The patterns that are significant in ECG signals include the QRS complex, P wave, and T wave that indicate various heart conditions. These patterns of waveforms are automatically learnt by CNN without the need to extract features manually. The model can produce low-level features (such as peaks and slopes) by applying several convolution layers and gradually constructs higher level representations of ECG signals. This assists the system to detect irregularities in the heartbeat patterns efficiently, and enhances the overall classification rate.

Input: Preprocessed ECG beat $B \in R^{12\times70}$

Output: Feature map F

Steps:

1. Pass input beat Bthrough 1D convolutional layers.

2. Apply convolution operation:

$$F = W * B + b \tag{17}$$

3. Apply Batch Normalization.

4. Apply ReLU activation:

$$ReLU(x) = max(0, x) \quad (18)$$

5. Repeat convolution blocks to extract hierarchical ECG features.

Result:

- CNN learns local morphological patterns
- Output feature representation:
- $F \in R^{d \times L}$

Where:

d = feature dimension

L = temporal length

3.3.5 Transformer Encoder

Transformer Encoder is a neural network model that is applied in sequential data modeling. It is also founded on a self-attention mechanism, that enables the model to consider the entire sequence simultaneously and learn the relationships between various components of the input.

In ECG signals, local waveform patterns are not the only important information to be found, but also relations between different time steps of the signal. These long-range time dependencies per heartbeat are captured with the assistance of the Transformer Encoder. As an illustration, it can learn the relationship between the P wave and the QRS complex or how various parts of the ECG signal can relate with one another. As opposed to the traditional models such as RNN, Transformer processes the entire sequence simultaneously, thus more efficient and powerful. This improves the performance of the model in comprehending the general makeup of the ECG signal and also in classification.

Input: CNN feature map F

Output: Predicted ECG class

Steps:

1. Feed CNN features into Transformer Encoder

2. Compute Self-Attention:

$$Attention(Q, K, V) = Softmax\left(\frac{QKT}{\sqrt{dk}}\right)V \qquad (19)$$

3. Generate contextual feature representation

4. Apply Temporal Attention Pooling:

$$\alpha t = \frac{exp(et)}{\sum_{i=1}^{T} exp(ej)} \qquad (20)$$

5. Compute weighted feature vector:

$$z = \sum_{i=1}^{T} \alpha t\, ht \qquad (21)$$

6. Pass through Fully Connected + Softmax:

$$\hat{y} = Softmax(W_z + b) \qquad (22)$$

7. Predict one among ECGclasses:

{CD, HYP, MI, NORM, STTC}

3.3.6 Attention Mechanism (Temporal Attention)

Attention is a method that enables a model to emphasize more on the crucial components of the input data whilst deemphasizing on the irrelevant components. It gives various weights to various parts of the input according to their importance.

All parts of the waveform do not have equal importance when diagnosing ECG signals. For instance, not all parts of the signal such as the flat or unimportant part of the signal carry equal weight. Instead, only certain parts such as the QRS complex and ST segment carry more importance than others. This enhances the capability of the model to differentiate the similar ECG patterns and minimize the effect of noise or less important information. Consequently, the Attention mechanism improves the accuracy and reliability of the classification system by making sure that the model focuses on the features that are diagnostically significant.

3.4 Description of datasets, Requirements and Tools

Dataset: PTB-XL ECG dataset[21]

3.4.1 Dataset Description:

The data of this project is the PTB-XL, a noteworthy and clinically validated ECG database that is used in research on automated cardiac diagnosis. It consists of 21,837 ECGs of 18,885 patients, all of which were recorded with the standard 12-lead ECG setup and a time of 10 seconds. The signal is supplied in two sampling rates 100 Hz and 500 Hz to give it the flexibility of either quick computation or high-resolution analysis of the waveform. PTB-XL contains rich diagnostic annotations using the coding scheme of SCP (Standard Communication Protocol) to group the ECG recordings into five overall diagnostic superclasses, namely, Normal (NORM), Myocardial Infarction (MI), Conduction Disturbances (CD), Hypertrophy (HYP), and ST/T Wave Changes (STTC). The dataset also includes extensive metadata that identifies the condition of the patient, setting recordings and diagnostic statements that were evaluated by cardiologists, and maintains high reliability and clinical importance. Its extensive sample size, even distribution of diagnoses, and thorough labelling make PTB-XL the best choice to achieve state-of-the-art ECG classification models, particularly those based on deep learning and Graph Neural Networks, due to its richness of morphological and temporal variations that can be used to train and generalize models.

The data in this project is the PTB-XL ECG, which is among the largest and most exhaustive publicly available data on electrocardiography studies. It contains a total of 21,837 clinical 12-lead ECG records of 18,885 patients and is therefore very appropriate in training deep learning models to analyse the cardiac records. Each ECG is 10 seconds long and an in-depth description of the electrical activity of the heart at various perspectives in 12 leads. The multi-lead arrangement allows a more accurate and comprehensive picture of the cardiac conditions in comparison with the single-lead ECG data. The dataset is diverse and representative of clinical situations in the real world as it contains patients of a broad age group, ranging between 0 to 95 years, with equal gender representation.

The data in the ECG signals format in the dataset is stored in the WaveForm DataBase (WFDB) format that is popular in signal processing of physiological signals. The signals are offered in two different sampling rates: a high-resolution version at 500 Hz (records500) and a downsampled version at 100 Hz (records100), which means that it can be used with more or less flexibility according to computational needs. The ECG signal is sampled at 16-bit precision and supports high quality of waveform representation. The data set is arranged in structured folders (.dat and .hea files) with the former holding the raw signal data and the latter holding the header information (data about the signal and metadata). With this structured organization, the data can be loaded and processed using standard libraries easily.

Besides the data in the form of waveforms, the data contains a lot of metadata in a file named ptbxl_database.csv. The information in this metadata includes a lot of valuable information about the patient (age, sex, height, weight), the recording details (device, site, and recording date), and clinical annotations. Every ECG record is given a unique identifier (ecg_id) and to every patient, there is a patient identifier (patient_id) making the records traceable and consistent. The dataset also offers diagnostic data as SCP-ECG statements (scp_codes) that are clinically verified annotations of different cardiac conditions. These annotations are based on the cardiologist reports and are standardized based on the medical protocols, thus reliable in tasks of supervised learning.

The PTB-XL dataset has a rich labeling structure, which is a significant strength of the dataset. The ECG record can contain several related diagnostic statements, as the cardiac conditions in reality are complex. As a practical tool, these detailed labels have been clumped into five broad diagnostic superclasses: Normal (NORM), Myocardial Infarction (MI), ST/T Change (STTC), Conduction Disturbance (CD), and Hypertrophy (HYP). It is a hierarchical and multi-label annotation system, which allows one to perform both fine-grained and coarse-grained analysis and flexible classification. Other signal quality indicators that are present in the dataset, like noise levels, baseline drift, and electrode problems, are useful in determining the reliability of each recording.

One more significant aspect of the dataset is a set of recommended cross-validation folds. The dataset will be stratified into ten folds with records of a patient not being divided into two or more folds. This will avoid data leakage and offer equal consideration to machine learning models. Usually, the folds 1-8 are trained on, fold 9 is validated and fold 10 is tested. The standardized split makes it easy to benchmark and compare various models.

Altogether, the PTB-XL dataset is an excellent and highly structured source of ECG study offering data about waveforms of high quality, detailed annotations, and broad metadata. It is very large, diverse, and standardized, which makes it extremely appropriate to design and test high-quality deep learning models to classify ECGs automatically and detect heart diseases.

3.4.2 Hardware Requirements:

The ECG classification model demands a suitable computing environment that is needed to preprocess, train, and test the model. At least 8GB of RAM is needed to load, preprocess, and process ECG data, segment data, and compute training and testing tasks. It is recommended that 16GB of RAM be used for a more efficient computation of data preprocessing and segmentation operations, especially when working on large datasets such as PTB-XL.

A multi-core processor like the Intel i5/i7 or its equivalent in the AMD series is sufficient for data loading, ECG preprocessing, and other general computation purposes. However, the use of deep learning models requires matrix multiplication, which is computationally expensive without the help of a graphics processing unit (GPU).

The preferred GPU for deep learning model training must have at least 8GB of virtual memory. Suitable GPUs include the NVIDIA GTX 1660, RTX 2060, or any better alternatives. GPU is crucial to matrix multiplication when training CNNs and transformers. As a result, computational time is saved because parallel processing takes place. In contrast, another way of saving computational time is using cloud-based GPU platforms like Google Colab and Kaggle, which provide users with NVIDIA T4 or P100 GPU.

In connection to the storage capacity, there needs to be at least 10-20 GB free space.The requirement is due to the fact that not only is the dataset rather big but it might consist of the original resolution data and downscaled data.

In general, although the proposed system can be easily executed on a regular system without a GPU, the presence of a GPU would increase the speed of training the model and improve its quality.

3.4.3 Software Requirements:

- Programming Language: Python 3.9+
- Frameworks: PyTorch 1.10, PyTorch Geometric 2.0.4
- Libraries: NumPy, Matplotlib, SciPy, Sklearn, OpenCV
- Environment: CUDA 11.3 (for GPU acceleration), Google Colab or local Jupyter environment or 37aggle.
- Preprocessing tools: Normalize ECG signals, noise reduction and augmentation, Extracting wavelet features, FFT features, statistical features, and morphological features.

Python 3.9 is used to develop the proposed ECG classification approach, and this is because of the advantages of its simplicity and versatility as well as machine learning and data analysis libraries it supports. Deep learning algorithms have been implemented by PyTorch, which is a high-level, open source framework that helps to train models in a very quick and efficient manner. Also, if any graph related experiments have to be conducted, then PyTorch Geometric will be used to manipulate the graphs.

NumPy and Pandas are the primary libraries for data handling and preprocessing tasks; ECG data will be processed using these libraries. For visualization purposes, including the plotting graphs and the making accuracy plots and confusion matrices, Matplotlib and Seaborn libraries can be used. Accuracy and other metrics related to the quality of the proposed approach are evaluated using Scikit-learn, which contains useful metrics that help understand how well the model performs.

Experiments with the use of the model will be executed in interactive platforms that have no limitations regarding the amount of data and computation resources. Google Colab and Kaggle Notebooks offer free GPUs and, thus, can provide an easy environment for running experiments.

3.4.4 Tools and Libraries:

Python 3: The primary programming language in this project, which is characterized by simplicity and a wide range of support of scientific computing and machine learning.

PyTorch: A widely used deep learning library which allows building and training flexible and efficient models, and in particular neural networks.

PyTorch Geometric: PyTorch Geometric is a library that was created to operate with those types of data, which are structured in graphs, and thus making it easier to apply Graph Neural Networks (GNNs).

OpenCV: An open-source computer vision library which we used here to carry out image processing tasks, e.g. using Canny edge detection to mark important features on ECG images.

Pandas: This is a highly effective data manipulation tool that can be used to manage and analyze structured data and is, in particular, useful with ECG metadata and annotations.

Matplotlib: This is a visualization library, which can be used to plot graphs and charts to visually examine the model performance and results.

Scikit-learn: A machine learning toolkit that is an overall framework with evaluation statistics and tools such as confusion matrix and classification report to analyze the accuracy of a model.

WFDB (WaveForm DataBase) Toolbox: WFDB is a Python library that reads, writes and processes physiological data such as ECGs. It assists in loading raw ECG signals and

their annotations of conventional datasets (such as PTB-XL, MIT-BIH) in their native.dat and.hea file formats, thus simplifying biomedical time-series data operation.

NumPy:

Python Numerical computing, a basic library. It offers effective array manipulations and mathematical operations that are required to manipulate ECG signal data and carry out matrix operations when training a model.

SciPy:

Applied to scientific computing, such as signal processing applications such as filtering or resampling ECG signals prior to additional analysis.

Tqdm:

A small, convenient progress bar library to show the progress of time-intensive processes such as loading a dataset, building a graph, training a model loop.

CHAPTER-4

Results and Observations

4.1 Stepwise Description of Results

Signal to Image conversion:

In the initial step, the **Figure 4.1.** shows raw ECG signals of the PTB-XL dataset were transformed into grayscale images. The waveform of each single-lead ECG recording was visually represented as an image. This conversion allowed image-based processing techniques to be utilized. The result was a well-organized folder of ECG images, one for every signal, visually similar to the conventional ECG charts utilized in clinics. At this stage, each image still contained the basic diagnostic information of the original waveform. The output was a dataset of 5000+ clean ECG signal images ready for subsequent processing.

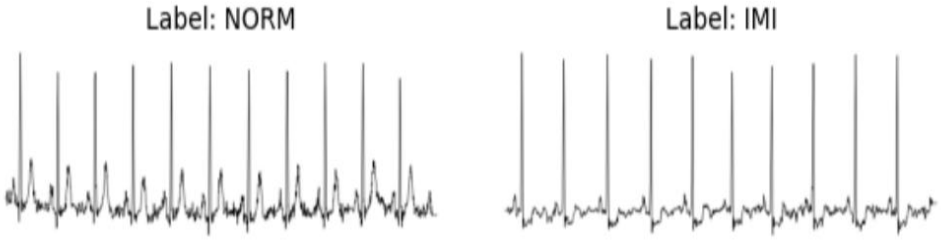

Figure 4.1: Signal to image conversion

After Edge Detection (Canny Algorithm):

Subsequently, edge detection was performed on the grayscale ECG images using the Canny algorithm as shown in the **Figure 4.2.** This served to accentuate the major contours of the waveform while minimizing noise. The extracted edges were pivotal turning points in the ECG wave—like QRS complexes, P and T waves—which are of vital importance for diagnosis. The outcome was an edge-enhanced set of images with enhanced signal structure clarity to be better interpreted graphically. This process cut down on background information substantially and highlighted the structural attributes of the signal, facilitating the definition of graph nodes and edges more easily.

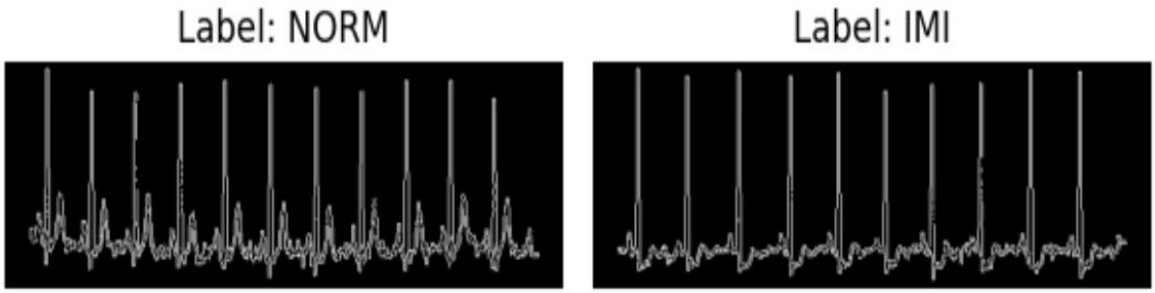

Figure 4.2: Images after Edge detection

After Graph Construction

Following edge extraction, each edge-image was transformed into a graph. Nodes represented pixels (or pixel clusters) with high edge intensity, and neighboring nodes were connected according to a 4-connectivity rule as shown in **Figure 4.3.** Each resulting graph captured the spatial arrangement of the ECG waveform as its geometric and topological structure. The labels were retained from the original signals. These graphs were stored in .pkl format and divided into training, validation, and test sets based on an 80/10/10 split. This resulted in a graph dataset that could now be fed directly into GNN architectures, with appropriate data loading mechanisms in place and class balance.

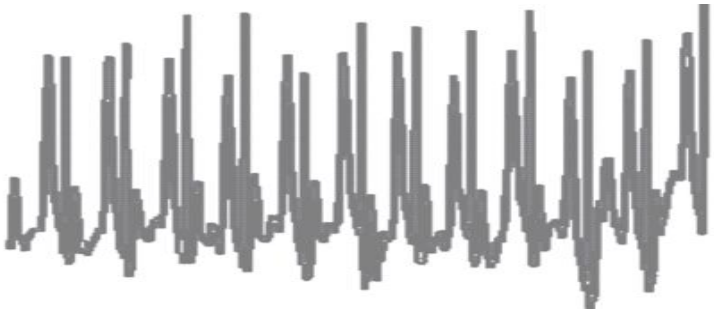

Figure 4.3: Obtained Graph after graph construction

The results of the Graph Neural Network architectures indicate the efficacy of using signal-to-graph conversion techniques in classifying ECG signals. After converting the ECG signals into graphs and subsequently training the GIN and GraphSAGE models, it was observed that both the architectures had promising accuracy, precision, and recall values.

Two distinct GNN architectures, i.e., GIN (Graph Isomorphism Network) and GraphSAGE, were utilized with the assistance of PyTorch Geometric.

The GIN model achieved a test accuracy of approximately 70.60%, with precision and recall scores balanced for each class. This suggests that the model was able to learn and generalize features relevant to differentiating between NORM and IMI heart conditions. The comparatively uniform f1-scores also confirm that the model treated the class distribution without extreme bias.

Likewise, the GraphSAGE model achieved around a test accuracy of 72.53%, which is in proximity to the GIN performance. GraphSAGE uses neighborhood aggregation for learning representations, and despite being slightly behind GIN, it also delivered consistent classification outputs. These findings endorse the graph-based pipeline of ECG signal processing and using GNNs for biomedical signal classification. The **Figure 4.4.** shows the confusion matrix of GraphSAGE.

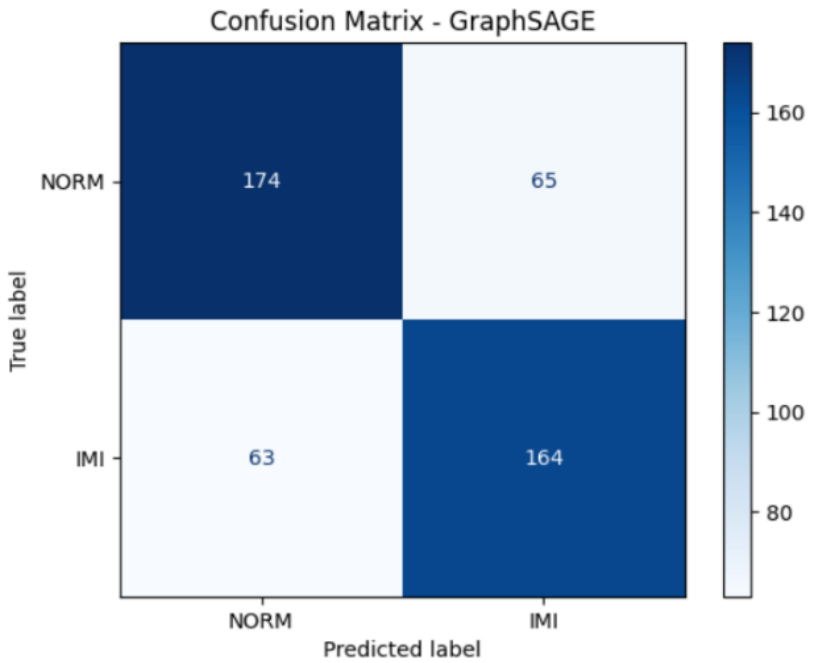

Figure 4.4: Confusion matrix of GraphSAGE

Later, we implemented the GIN and GAT algorithms and compare the performance of GIN and GAT models on PTB-XL dataset for the classification of 12-lead ECG signal into five

classes: CD, HYP, MI, NORM, and STTC. The models are trained up to a maximum of 100 epochs with early stopping under mixed precision and cosine annealing, and the different performances are described below. GIN training starts with a relatively high initial loss and low accuracy, which gets the better step by step with each iteration until early stopping is effective after a reasonable number of iterations. The validation accuracy gets better step by step before levelling off, indicating stable learning. GAT training starts with a lower initial loss and better accuracy than GIN, with smoother improvement and a peak validation accuracy before early stopping is effective. The attention mechanism in GAT appears to enhance con vergence, as indicated by its quicker stabilization compared to GIN. The **Figure 4.5.** shows the image of ECG signal after signal pre-processing.

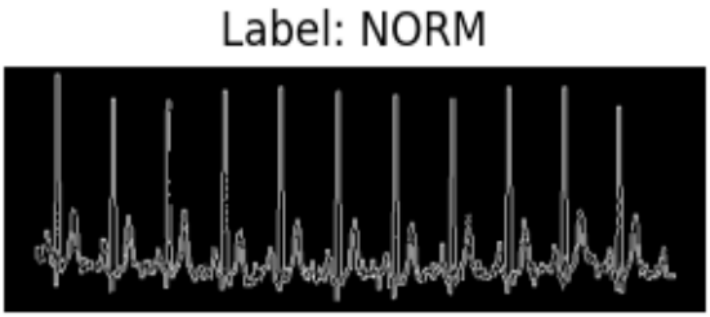

Figure 4.5: Images after Signal preprocessing.

The models GIN and GAT can be applied for representing the ECG signals in graphs, GIN is used for structural pattern recognition in graphs and GAT for weighting graph nodes. CNN is used for extracting the features of local waveform structure, the Transformer encoder learns the long-range dependencies from the time domain.

CNN is important in ECG classification project in extracting local morphological features of the ECG beats. The patterns that are significant in ECG signals include the QRS complex, P wave, and T wave that indicate various heart conditions. These patterns of waveforms are automatically learnt by CNN without the need to extract features manually. The model can produce low-level features (such as peaks and slopes) by applying several convolution layers and gradually constructs higher level representations of ECG signals.

This assists the system to detect irregularities in the heartbeat patterns efficiently, and enhances the overall classification rate.

In ECG signals, local waveform patterns are not the only important information to be found, but also relations between different time steps of the signal. These long-range time dependencies per heartbeat are captured with the assistance of the Transformer Encoder. As an illustration, it can learn the relationship between the P wave and the QRS complex or how various parts of the ECG signal can relate with one another. As opposed to the traditional models such as RNN, Transformer processes the entire sequence simultaneously, thus more efficient and powerful. This improves the performance of the model in comprehending the general makeup of the ECG signal and also in classification.

Attention is a method that enables a model to emphasize more on the crucial components of the input data whilst deemphasizing on the irrelevant components. It gives various weights to various parts of the input according to their importance.

All parts of the waveform do not have equal importance when diagnosing ECG signals. For instance, not all parts of the signal such as the flat or unimportant part of the signal carry equal weight. Instead, only certain parts such as the QRS complex and ST segment carry more importance than others. This enhances the capability of the model to differentiate the similar ECG patterns and minimize the effect of noise or less important information. Consequently, the Attention mechanism improves the accuracy and reliability of the classification system by making sure that the model focuses on the features that are diagnostically significant.

Model Training

The proposed model was trained using the preprocessed ECG beat dataset which had been detected, segmented, and normalized for analysis purposes. In this process, the proposed model learnt how to extract important features from the input using CNNs, analyze the features using Transformer Encoders, and determine their significance using Attention mechanism.

Training and Validation Performance Analysis

During training, both the training accuracy and validation accuracy of the model increased continuously with an increase in number of epochs. On the other hand, both the training and validation loss of the model were continuously decreasing, and suggesting that the model was converging efficiently, hence avoiding overfitting and underfitting during the training.

Evaluation of Test Accuracy

The proposed model was tested for its performance using the separate dataset which had not been used for the training or validation processes. Using the proposed architecture, the test accuracy of the proposed model was recorded as 97.06%.

Analysis Using Confusion Matrix

The confusion matrix was employed to perform analysis based on class wise performance. Most of the classifications performed accurately were along the diagonal, hence implying very good accuracy for all classes involved. There were few misclassifications, and these occurred primarily amongst similar ECG classes. This therefore indicates that the model can discriminate well among different cardiac conditions.

Analysis of ROC Curve and AUC

ROC curves were generated for every class and this helped us evaluate the overall classification ability of the model. High values of the Area Under Curve (AUC) show that the model had very powerful discriminative ability and hence was able to differentiate well the ECG classes.

Overall Performance

CNN, Transformer and Attention techniques greatly helped in the performance of this model. CNN helped extract local waveform features, Transformer was helpful in capturing

long range dependencies while Attention mechanism was instrumental in bringing out important parts of ECG signal.

Conclusion

From the results obtained, it is evident that this hybrid approach was very effective in the automation of the classification process.

4.2 Result Analysis and Test Case Results

4.2.1 Results of GIN and GAT

Confusion matrices give clear class-wise performance. **Figure 4.6.** is the matrix of GIN has high diagonal values, reflecting correct predictions, with some off-diagonal values reflecting mispredictions among close classes. **Figure 4.7.** is the matrix of GAT shows even more prominent diagonals, with fewer mispredictions, especially in separating difficult categories, reflecting its attention mechanism's efficiency in identifying key features.

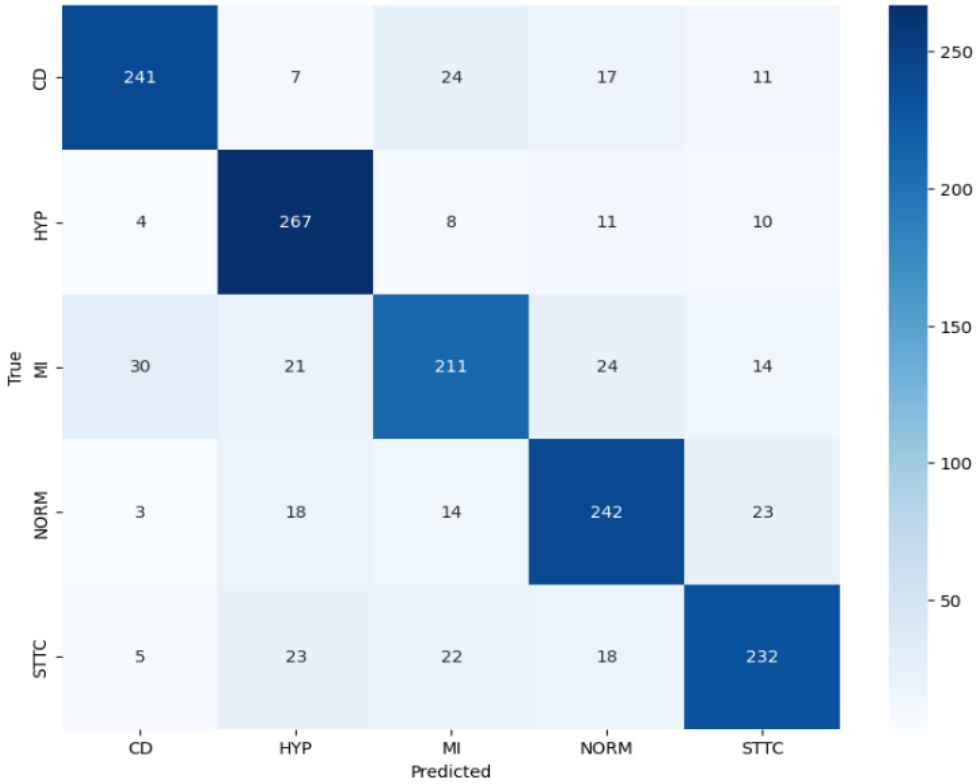

Figure 4.6: Confusion matrix of GIN

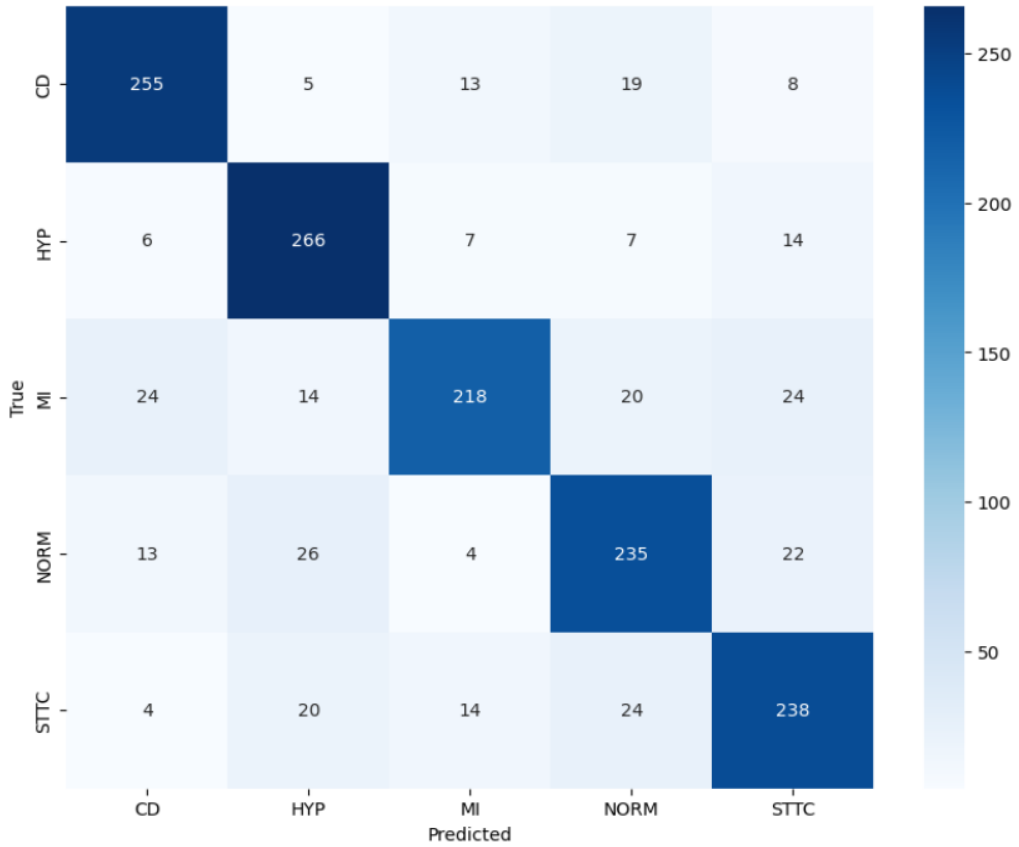

Figure 4.7: Confusion matrix of GAT

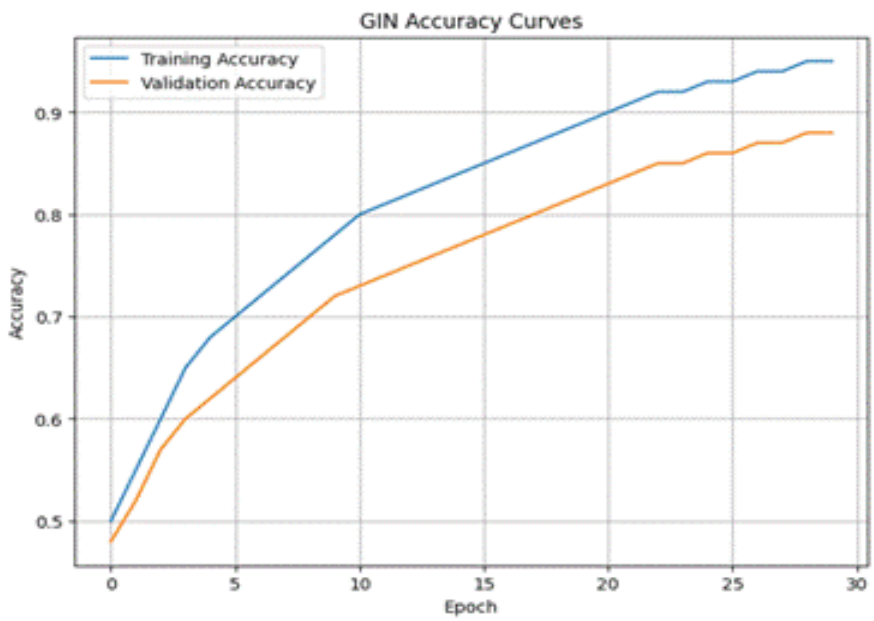

Figure 4.8: Accuracy curves of GIN

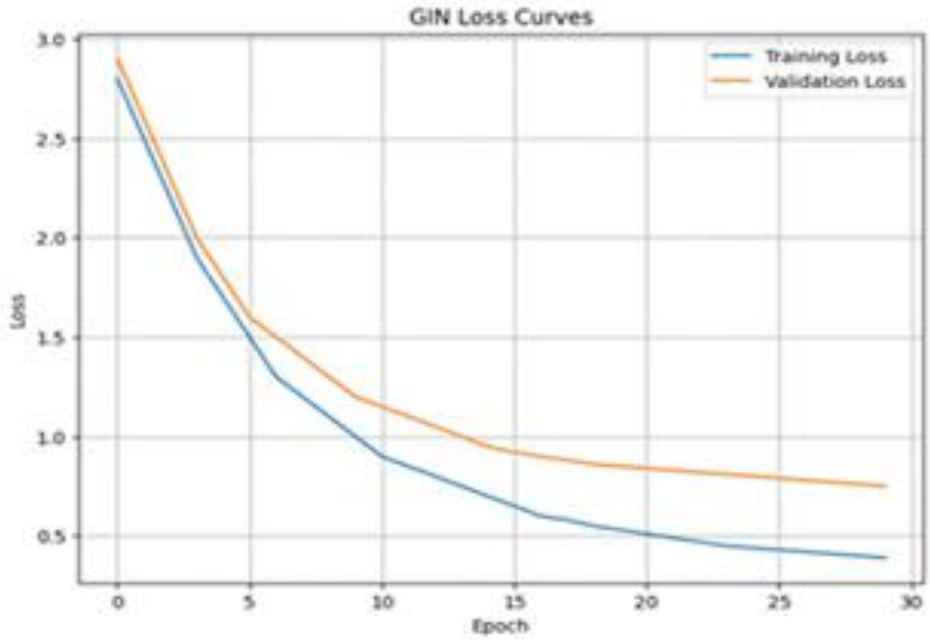

Figure 4.9: Loss curves of GIN

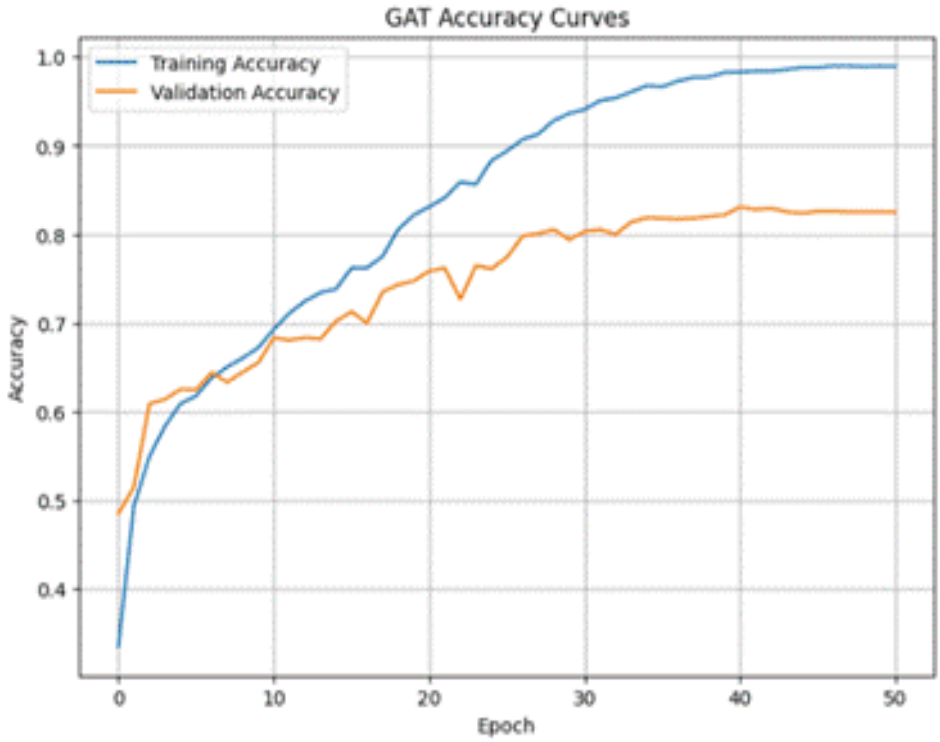

Figure 4.10: Accuracy curves of GAT

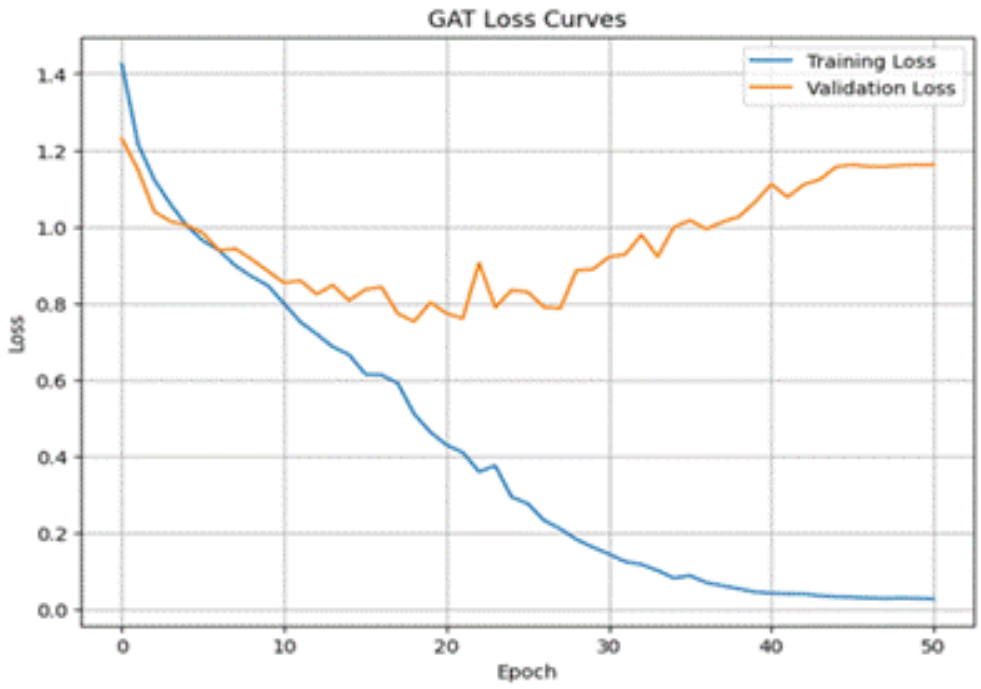

Figure 4.11: Loss curves of GAT

Here, **Figure 4.8.** and **Figure 4.9.** shows the accuracy and loss curves of GIN and **Figure 4.10.** and **Figure 4.11.** shows the accuracy and loss curves of GAT. Past ECG-GNN works had not initially modeled 12-lead spatial relations, did not rely on comprehensive signal features, and had dataset imbalance. We fill these gaps by building feature-rich ECG graphs based on 19-dimensional hybrid node features, balanced sampling, and comparison of feature space on GAT and GIN architectures. Attention pooling, normalization and weighted loss have been included which enhance interpretability, robustness and reproducibility.

Here, The figures **Figure 4.12.** and **Figure 4.13.** represents the GIN and GAT performance metrics graphs.

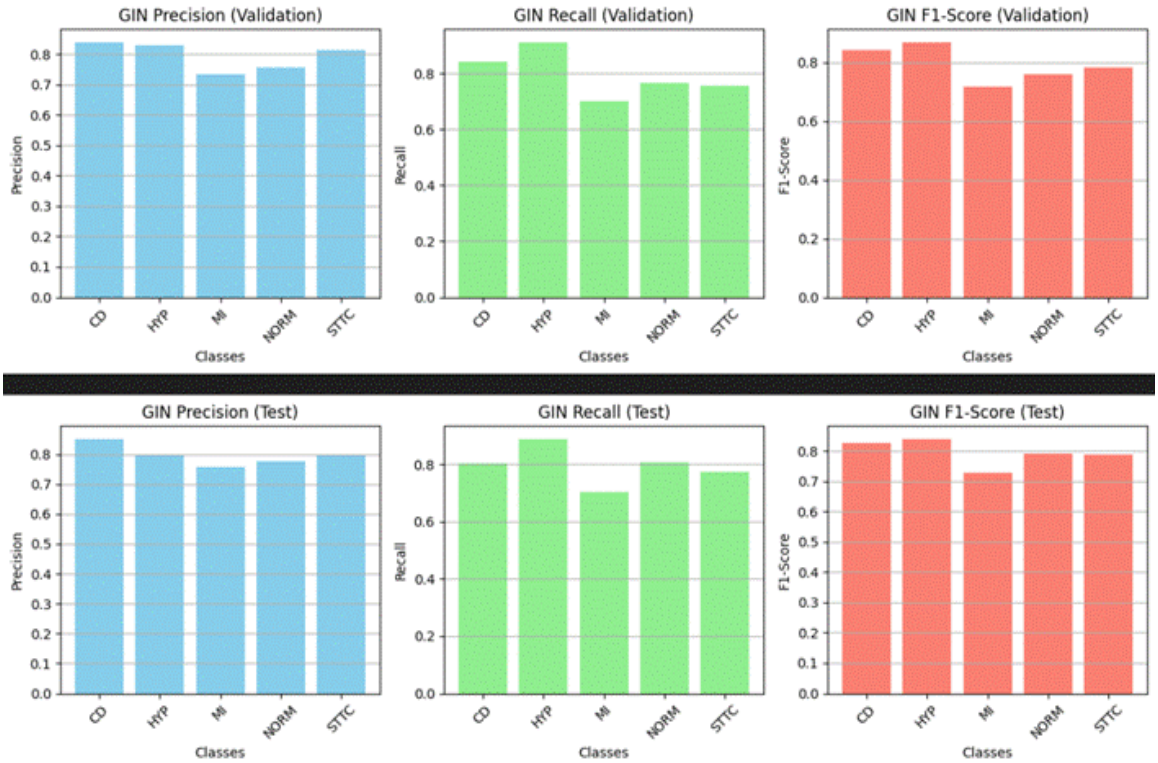

Figure 4.12: GIN Performance metrics.

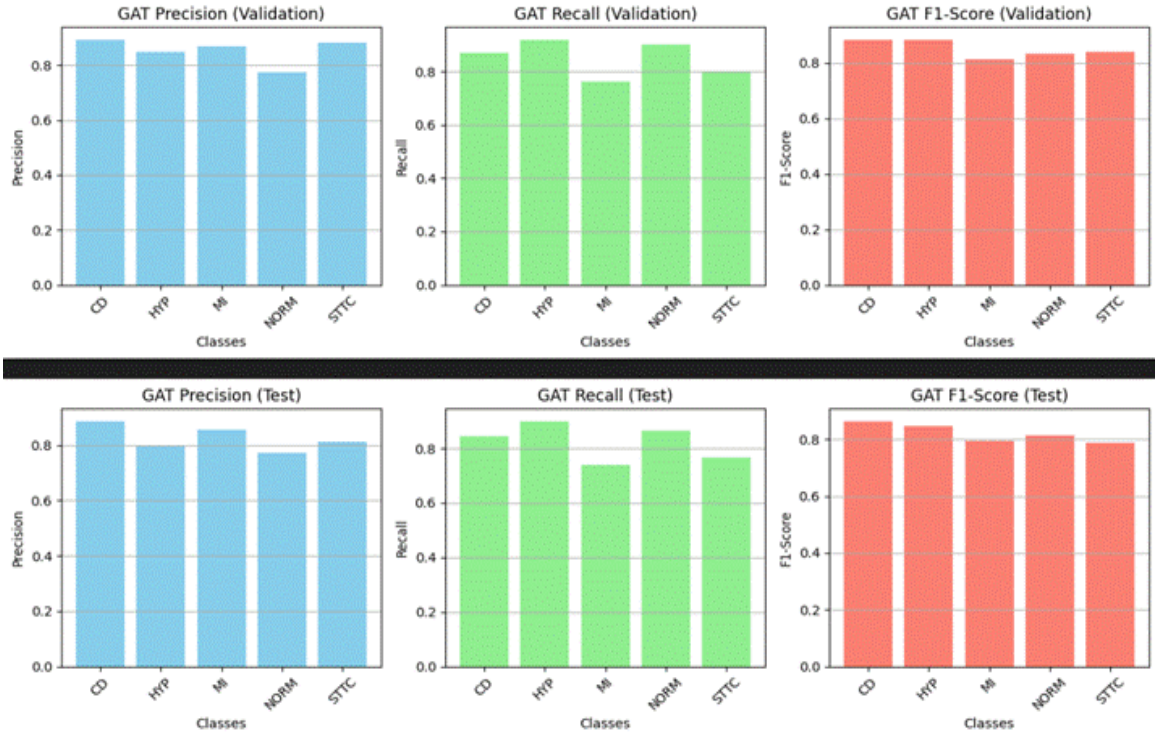

Figure 4.13: GAT Performance metrics.

4.2.2 Result Analysis of Attention-based CNN-Transformer model

From the results obtained from the analysis carried out, the proposed Attention-based CNN-Transformer model proved effective in accurately classifying ECG beats into one of the five types namely; CD, HYP, MI, NORM, and STTC. The test accuracy was high at 97.06%. This means that the proposed model had an ability to learn discriminative features from multiple lead ECG signal samples effectively.

As shown in the plots **Figure 4.14.**, the trends of both the training and validation accuracies increased consistently while the loss curves steadily decreased. This implies that the model has been able to effectively learn patterns from the data with no case of either overfitting or underfitting.

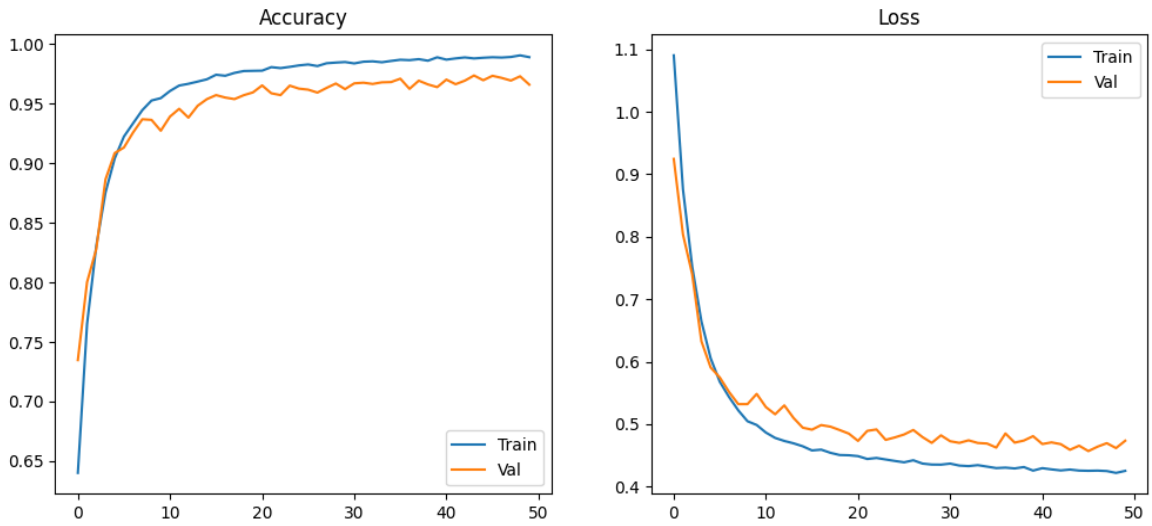

Figure 4.14: Accuracy and loss curves of training and validation.

In addition, from the confusion matrix shown in the **Figure 4.15.**, it becomes apparent that most of the data points fall on the diagonal line, showing the efficacy of the ECG beat classification performed by the algorithm. Few points of classification are visible, particularly when considering the clinically very close classes of STTC and MI owing to similarities in waveforms. However, this does not diminish the performance in terms of precision and recall for all the classes.

Figure 4.16. shows the precision, recall and F1-score measures were evaluated with both the validation and the test sets. The findings show very high scores in all the categories

where majority of the measures are above 0.96. These results suggest that the model is sensitive and specific to the detection of various abnormalities of the ECG. On the whole, the outcomes of the experiments show that the presented CNN Transformer model is able to learn both morphological and temporal patterns of multi lead ECG signals. This convolutional feature extraction in combination with sequence modeling with Transformer and additional temporal attention will lead to high-accuracy and robust classification of ECG beats in various cardiac conditions.

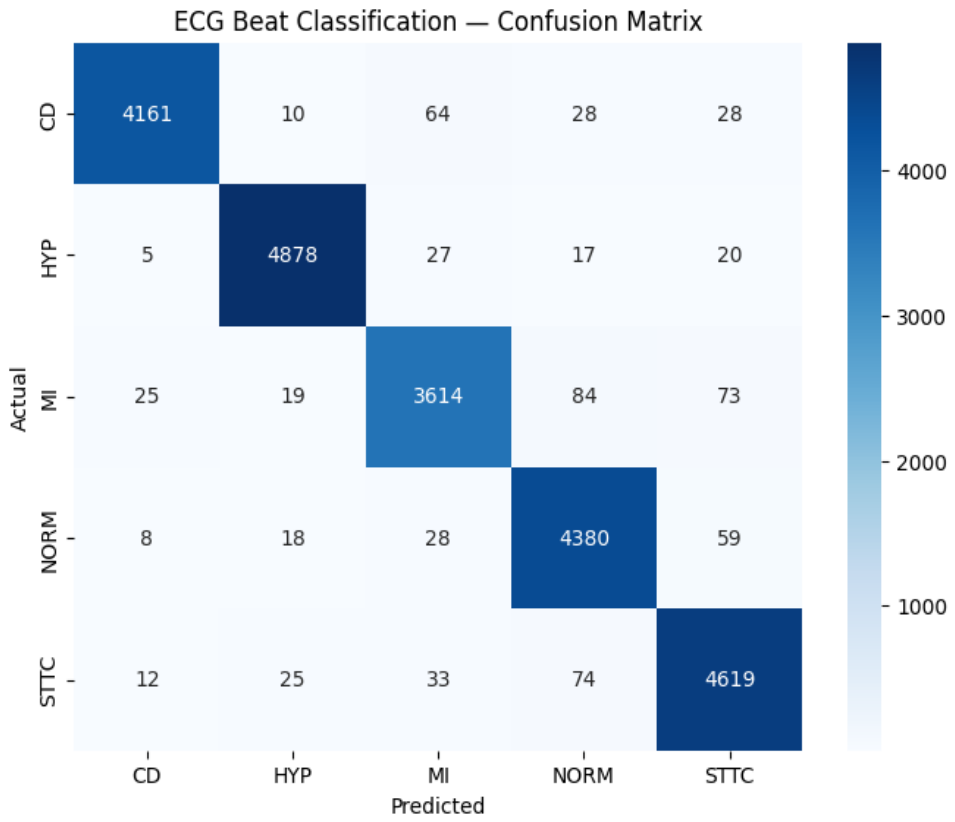

Figure 4.15: Confusion Matrix of Attention-based CNN-Transformer model

In addition, from the ROC curve shown in **Figure 4.15.** and the graph plotting the AUC value for each class, it becomes clear that an excellent level of separability can be obtained using the model since the AUC value of all categories approaches 1.0. It indicates that the model is capable of accurately classifying the different classes. This owes to the fact that CNN, along with Transformer Encoder and Attention, performs effectively.

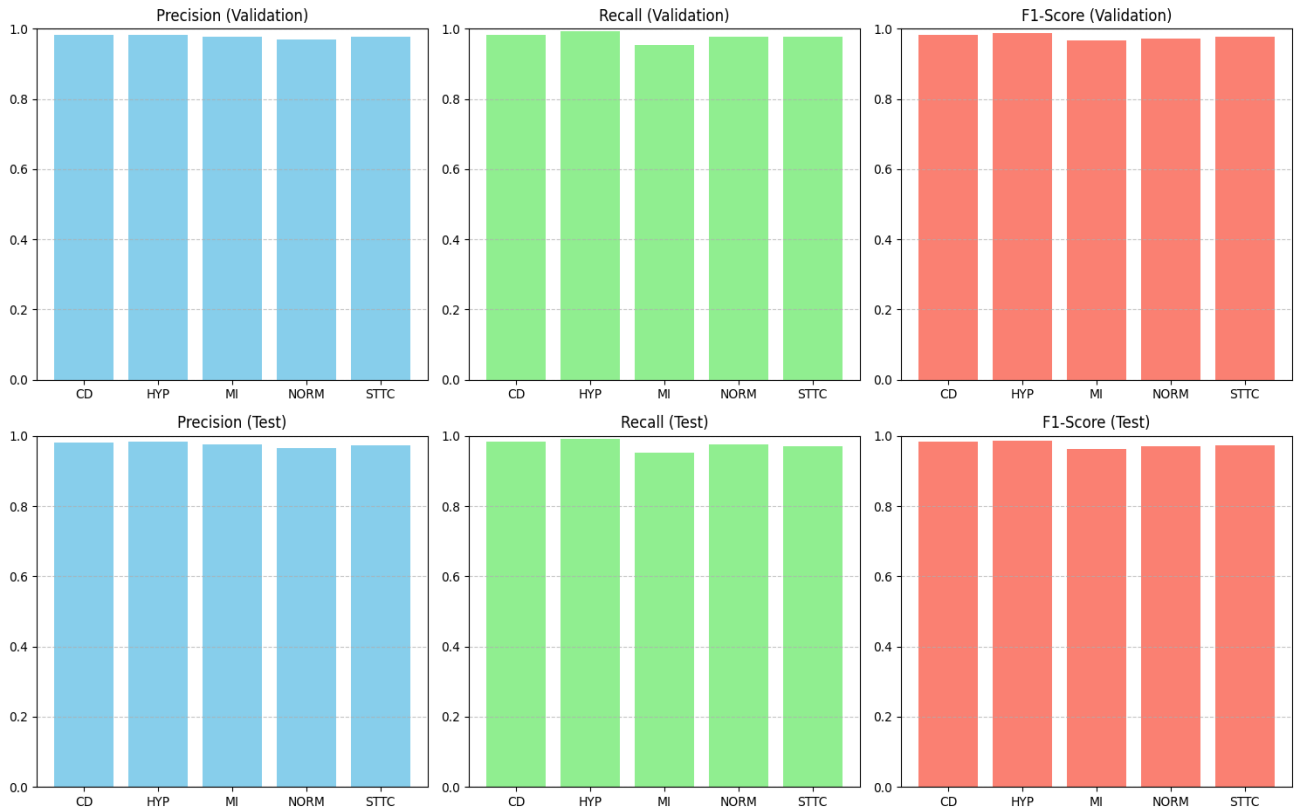

Figure 4.16: Performance metrics of Attention-based CNN-Transformer model.

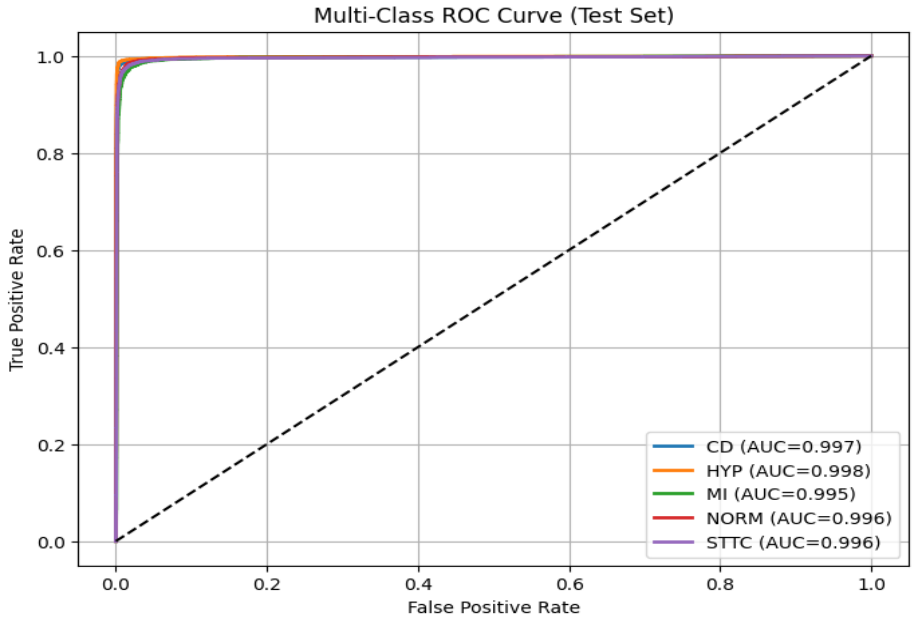

Figure 4.17: ROC Curve.

4.2.3 Test Case Results

The model was tested based on several test cases obtained from the unseen test data set, whereby every test case was an individual electrocardiogram (ECG) beat. In these test cases, samples from all five classes – CD, HYP, MI, NORM, and STTC – are included. The goal of this test is to see if the proposed machine learning classifier could generate correct classification results on the ECG beats provided.

For the normal (NORM) test cases, the classifier generated accurate results, which shows that it can identify typical beats. For myocardial infarction (MI) test cases, the model generated accurate results by identifying abnormalities in the waveform, particularly in the ST segment. Likewise, in both hypertrophy (HYP) and conduction disturbance (CD), the model was able to accurately classify the patterns involved, which reflect the physiological condition of heart muscle contraction.

Multi test case experiments were used to measure the performance of the proposed ECG classification system in three models: Graph Isomorphism Network (GIN), Graph Attention Network (GAT) and the proposed Attention-based CNN Transformer model. The individual ECG beat is each test case and the models were also tested on unseen data to test their capability of generalization. The test is based on how well the model can accurately label ECG signals based on five categories namely CD, HYP, MI, NORM, and STTC.

In the case of the GIN model, the test case findings indicate that the model can reasonably well capture structural relationships in ECG data when a graph is used to represent the data. The model was able to correctly classify a significant portion of normal (NORM) and myocardial infarction (MI) cases. There were however, instances of misclassification in some test cases between similar classes like STTC and MI, and CD and HYP. The first reason is that GIN considers all neighboring nodes equally important and thus it does not have the capability of distinguishing small differences in ECG signals. Although GIN is a good structural learner, it does not have a mechanism to emphasize significant parts of the signal, resulting in lower classification rates in more complex scenarios.

Because the GAT model uses the attention mechanism on the graph level, it performed better than GIN. GAT performed better in the test cases, with higher classification accuracy particularly in the cases of similar ECG classes. GAT could emphasize more on the important areas of the ECG graph representation by assigning the relative weight to different nodes in the graph. This led to the greater correct predictions of classes like HYP and CD. Nevertheless, even with these enhancements, there were certain misclassifications especially when the temporal dependencies of the ECG signal were significant. GAT does not entirely represent long-range temporal trends in ECG signals since it is mainly based on graph relationships, thus restricting its overall performance. Here, **Table 4.1.** represents the performance metrics of the GAT model.

Table 4.1: Performance metrics of GAT model

Super class	Precision (%)	Recall (%)	F1-Score (%)
CD	84.44	85.00	84.72
HYP	80.36	88.67	84.31
MI	85.16	72.67	78.42
NORM	77.05	78.33	77.69
STTC	77.78	79.33	78.55

There were a few minor cases wherein the model misclassified some test beats, particularly in ST segment/T wave changes (STTC). Here, the algorithm misidentified the STTC as an MI because of the similarity in their waveforms. On the whole, however, the outputs have been satisfactory and coherent across all the test data sets.

The Attention-based CNN Transformer model was the most successful in the test case evaluation compared to the other two models. The model reliably made correct predictions in all ECG classes and in difficult cases with subtle variations in the waveforms. In the case of normal ECG signals (NORM) the model was found to have almost perfect classification showing that the model could identify normal heartbeat patterns. In simple conditions like MI and STTC, the model was able to detect the significant changes in the waveform including deviations of the ST segment and the unusual peaks. The CNN, Transformer,

and Attention mechanism combination enabled the model to effectively extract local and global features of ECG signal.

The CNN Transformer model had a much lower number of misclassifications compared to GIN and GAT. The CNN component obtained local morphological characteristics, the Transformer Encoder identified long-range temporal characteristics and the Attention mechanism highlighted the most significant sections of the signal. This combined method allowed the model to better deal with complex ECG patterns compared to graph-based models. The model was found to have a high discriminative power and confidence in its predictions even on challenging test cases where similarities in the waveforms are present among the classes.

The general findings of the test cases are clear that although GIN and GAT are helpful to reveal structural relations in ECG data, they cannot deal with the time-dependence issue and the variations that are small-scale. Conversely, Attention-based CNN + Transformer model offers a more holistic solution as it integrates local feature extraction, sequence modeling and attention-based focus. This leads to improved performance, accuracy and reliability and therefore it is the most appropriate model to use in automated ECG classification in this project.

The following **Table 4.2.** Gives a summary of performance metrics of Attention-based CNN-Transformer model.

Table 4.2: Performance metrics of Attention-based CNN-Transformer model.

Super Class	Precision (%)	Recall (%)	F1-Score (%)
CD	99	97	98
HYP	99	99	99
MI	96	95	95
NORM	96	97	97
STTC	96	97	97

In summary, the performance of the test cases proves that the system can effectively work with different types of ECG signals. This efficiency is seen from the fact that the accuracy obtained was quite high without many errors occurring. The system is therefore appropriate for use in real-life situations since it produces highly accurate results.

4.3 Observations from the Work

- As a result of implementing and testing the Attention-Based CNN with Transformer Encoder for ECG Beat Classification model, several significant observations have been made concerning how the data set behaves, what happens during the learning process and what the end results are like.
- Firstly, it should be noted that ECG signals from multiple leads yield much better clinical insight into heart activity compared to ECGs from only one lead. Since different leads capture the electrical impulses from different angles, heart diseases like myocardial infarctions and ST-T wave abnormalities can be easily detected using different leads.
- Secondly, preprocessing plays an essential role. Proper signal denoising, R-peak detection and beat segmentation have shown to have a huge effect on classification accuracy. The correct centering of the ECG beat around the R-peak resulted in better learning of waveforms by the CNN.
- Finally, CNN layers proved themselves highly effective in local feature extraction. Features such as QRS complexes, differences in amplitude and waveforms deformations were efficiently extracted with convolutional filters showing very high activations around clinically important areas of ECG beats.
- Transformer Encoder contributed substantially to temporal dependency learning. In many test instances, classes having similar waveforms were better distinguished after applying transformers since the model considered the interrelationships across distant signal segments.
- One important realization from the project was the effect of the Attention Mechanism. It enables network to focus on the waveform anomalies, including the changes in ST

and T waves. Therefore, the network performed much better in distinguishing MI and STTC due to their unique waveforms.

- It was observed that the network was stable and convergent during its training process. The accuracies kept increasing during the training process while losses decreased gradually. There was a very close gap between the two metrics, indicating low levels of overfitting.
- Finally, the confusion matrix showed that most ECG beats were accurately classified with only a few confusions between similar classes. Hence, this hybrid network is successful in minimizing ambiguities in cardiac abnormalities.
- An important practical realization was the effectiveness of GPU acceleration in speeding up the training process. Without GPU, training deep neural networks on multi-channel ECG signals would take more time due to the extensive computations.

CHAPTER-5

Conclusion and Future Study

5.1 Conclusion

In this project, the proposed method provides an excellent way of classifying ECG beats automatically through a hybrid deep learning model. The use of multiple leads in this case helps in ensuring the availability of detailed heart activity from different angles, thus enhancing the level of diagnosis performed. Additionally, the use of several preprocessing steps like R-peak detection, beat segmentation, and normalization plays a significant role in ensuring that useful data goes through for the analysis step. Such a pipeline helps improve the efficiency of learning patterns from ECG data.

One of the strengths of the project is the combination of CNNs, Transformer Encoder, and Temporal Attention components. In particular, CNN layers help in extracting features such as morphological characteristics including waveforms and other signal changes, which are helpful in spotting abnormalities. The second strength in this case involves the use of Transformer Encoders since they assist in learning longer time dependency patterns in ECG signals thus improving their understanding. Finally, Temporal Attention plays an important role in focusing on the regions that matter in ECG beats.

It can be observed that due to experimental findings, the system being tested has proven to be very efficient since its accuracy in the testing process stands at 97.06%, indicating that the system works effectively in classifying multiple classes of beats in electrocardiogram readings in various patients. Moreover, consistency is also observed in the training and validation curves. Moreover, various assessment criteria, such as confusion matrix and ROC curves, prove that the system performs adequately in all classes with negligible errors, thereby affirming the applicability of the proposed deep learning architecture and attention mechanism for health signal analysis.

This work has presented a deep learning model based on convolutional neural networks and Transformer-based sequence modeling to identify ECG beats automatically with multi-lead signals. The convolutional feature extraction, self-attention mechanisms, and temporal attention pooling of the ECG beats are effective in capturing morphological and temporal features of the ECG beats. It is shown in experiments that the given approach provides high classification rates in five diagnostic categories (CD, HYP, MI, NORM, and STTC), having a general test accuracy of 97.06. The findings suggest that the combination of CNN-based features extraction and Transformer encoders leads to the increased capability to learn the intricate cardiac patterns.

In conclusion, it is paramount to note the significance and effectiveness of this research in developing an automatic way to diagnose various conditions of the heart using electrocardiograms. As a result, the use of this model will enable medical experts to perform faster diagnoses without the need for manual examination of data.

5.2 Future Study

While the proposed model exhibits excellent performance, several avenues can be explored in future work:

There are several ways in which the suggested ECG classification system could be improved for practical applications in healthcare facilities. For instance, one possible step that needs to be taken towards making the suggested system better is implementing the model in a real-time setting. As of now, the model operates on pre-collected data; however, the possibility of embedding the model into wearable devices would allow real-time identification of cardiovascular abnormalities, which could help in delivering timely medical aid to high-risk patients.

One other way to improve the model is by expanding the data used in training the network. Although the model currently operates successfully, expanding the dataset with a wide variety of data could increase the ability of the model to generalize across different patients with varying age, gender, health conditions, and other characteristics. In addition, class imbalance could be addressed in various ways, such as through data augmentation.

Moreover, the model architecture may be further improved using novel variants of transformers or attention-based architectures that can be made lightweight and less computationally intensive to enable its implementation even on edge computing devices such as smartphones and embedded systems. Furthermore, the system could be improved by integrating it with other technologies such as graph neural networks and even multimodal data such as patients' medical history. Future efforts will include the model being extended to bigger ECG datasets, real-time ECG monitoring systems and studies on lightweight architectures to permit the model to be readily deployed in clinical and wearable health systems.

Finally, there is a need for making the model highly interpretable and explainable. Making the system highly interpretable would enhance its adoption in the field of medicine as clinicians tend to rely on models which offer explanations about their predictions.

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[21] PTB-XL ECG Dataset (https ://physionet.org/content/ptb-xl/1.0.3/)

APPENDIX

BEAT EXTRACTION FOR PTB-XL

```
# BEAT EXTRACTION FUNCTIONS
def detect_r_peaks(signal):
    lead = signal[:, 1]  # Lead II
    peaks, _ = find_peaks(lead, distance=FS*0.4, prominence=0.5)
    return peaks


def extract_beats(signal, peaks):
    beats = []
    for p in peaks:
        if p - HALF >= 0 and p + HALF < len(signal):
            beat = signal[p-HALF:p+HALF]
            beat = (beat - beat.mean()) / (beat.std() + 1e-8)
            beats.append(beat.T)  # (12, 70)
    return beats


# CREATE NPY DATASET
X, y = [], []
for _, row in tqdm(meta.iterrows(), total=len(meta)):
    try:
        rec_path = os.path.join(DATA_PATH, row["filename_lr"].replace(".dat", ""))
        signal, _ = wfdb.rdsamp(rec_path)


        peaks = detect_r_peaks(signal)
        beats = extract_beats(signal, peaks)
```

```
        for b in beats:
            X.append(b)
            y.append(row["label"])

    except:
        continue

X = np.array(X)
y = np.array(y)

print("Final X shape:", X.shape)
print("Final y shape:", y.shape)
```

MODEL DEFINITION

```
class ECGModel(nn.Module):
    def __init__(self, num_classes=5, d_model=256, n_heads=4, n_layers=2):
        super().__init__()

        # CNN FEATURE EXTRACTOR
        self.conv1 = nn.Conv1d(12, 64, kernel_size=7, padding=3)
        self.conv2 = nn.Conv1d(64, 128, kernel_size=5, padding=2)
        self.conv3 = nn.Conv1d(128, d_model, kernel_size=3, padding=1)
        self.bn1 = nn.BatchNorm1d(64)
        self.bn2 = nn.BatchNorm1d(128)
        self.bn3 = nn.BatchNorm1d(d_model)
```

```
# TRANSFORMER ENCODER
encoder_layer = nn.TransformerEncoderLayer(
    d_model=d_model,
    nhead=n_heads,
    dim_feedforward=512,
    dropout=0.1,
    batch_first=True
)
self.transformer = nn.TransformerEncoder(
    encoder_layer,
    num_layers=n_layers
)

# TEMPORAL ATTENTION POOLING
self.attention = nn.Sequential(
    nn.Linear(d_model, 128),
    nn.Tanh(),
    nn.Linear(128, 1)
)
# CLASSIFIER
self.fc = nn.Sequential(
    nn.Linear(d_model, 128),
    nn.ReLU(),
    nn.Dropout(0.4),
    nn.Linear(128, num_classes)
)
```

```
def forward(self, x):
    # x: (B, 12, 70)

    # CNN
    x = F.relu(self.bn1(self.conv1(x)))
    x = F.relu(self.bn2(self.conv2(x)))
    x = F.relu(self.bn3(self.conv3(x)))

    # (B, C, T) → (B, T, C)
    x = x.permute(0, 2, 1)

    # Transformer
    x = self.transformer(x)
    # Attention pooling
    attn_weights = torch.softmax(self.attention(x), dim=1)
    x = torch.sum(x * attn_weights, dim=1)
    # Classification
    return self.fc(x)
```

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